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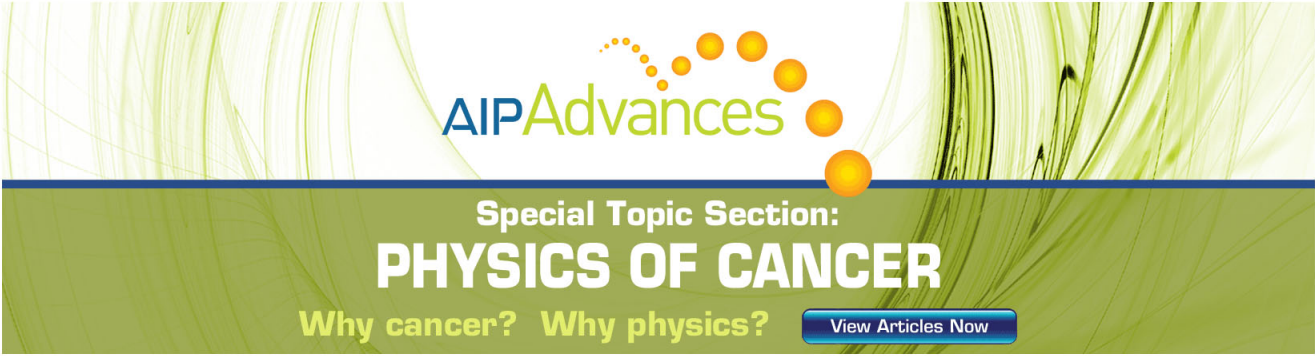
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Computer simulation studies of anisotropic systems. XXX. The phase behavior and structure of a Gay–Berne mesogen

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The Gay–Berne potential is proving to be a valuable model with which to investigate the behavior of liquid crystals using computer simulation techniques. The potential contains four independent parameters which control the anisotropy in the attractive and repulsive interactions. The choice of these parameters is not straightforward and it would seem that those employed in some simulations are not strictly appropriate for mesogenic rodlike molecules. Here we report a detailed computer simulation study of Gay–Berne particles interacting via a potential parametrized to reflect the anisotropic forces based on a fit to a realistic mesogenic molecule. The behavior of the phases and the transitions between them have been investigated for a system of 2000 particles using isothermal–isobaric Monte Carlo simulations. At low pressures, this Gay–Berne mesogen exhibits isotropic, smectic A and smectic B phases but, as the pressure is increased, so a nematic phase is added to the sequence. The nature of the phase transitions and the phase diagram are compared where possible with those of real mesogens. The structures of the four phases have been investigated in detail for a larger system of 16 000 particles using canonical molecular dynamics simulations at state points taken from the phase diagram determined from the Monte Carlo simulations. A wide range of singlet and pair distribution functions were evaluated together with orientational correlation coefficients and, for the smectic phases, a bond orientation correlation function. The results for these properties were used to identify the phases, to consider their structure at a quantitative level and, where possible, to make contact with experimental studies and the predictions of theories of liquid crystals. It would appear that with this parametrization, the Gay–Berne potential provides a powerful tool with which to understand the behavior of real liquid crystals and to test the predictions of theory. © 1999 American Institute of Physics. [S0021-9606(99)52014-X]

I. INTRODUCTION

Since the first computer simulation study of a model liquid crystal by Lebwohl and Lasher¹ in 1972, the powerful techniques of molecular dynamics and Monte Carlo simulation have proved to be of increasing importance in understanding the behavior of liquid crystals, especially at the molecular level. An essential feature of any simulation study is the interparticle potential and, for liquid crystals, four broad classes have been developed;² these are lattice systems, hard particles, single site soft potentials and atomistic models. The choice of the model to be used in a simulation is determined by a range of factors. This often includes a compromise between the computational resources available and the reality of the model, which is usually related to its computational complexity. In addition, the nature of the model can depend on whether the problem of interest is related to the behavior of a specific liquid crystal compound or the general behavior of liquid crystals. If the latter, then it is often important to be able to relate the phases formed and their properties to the essential molecular factors responsible for the

existence of liquid crystals. It is now agreed that these factors include the anisotropy in both the shape of the molecules and the attractive forces between them. These two factors are available within single site potentials where the parameters in the potential depend on the orientations of the two particles as well as that of the vector linking them; such models usually have the additional advantage of computational simplicity over an atomistic model for a specific mesogen.

One of the most useful single site or Corner³ potentials is that developed by Gay and Berne⁴ from the Gaussian overlap model proposed by Berne and Pechukas.⁵ The potential contains four parameters (κ , κ' , μ , ν) which determine the anisotropy in the repulsive and attractive forces, in addition to two parameters (σ_0 , ϵ_0) which scale the distance and energy, respectively (see Appendix A). The ratio of the end-to-end and side-by-side contact distances, κ , is related to the anisotropy of the repulsive forces but it also determines the difference in the depth of the attractive well between the side-by-side and the cross configurations; these quantities influence the ability of the system to form an orientationally ordered phase. The parameter κ' is the ratio of the well depths for the side-by-side and end-to-end configurations and it is this parameter which determines the tendency of the system to form a smectic phase. There are two further parameters, μ and ν , which influence the nematic and smectic forming

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character of the anisotropic attractive forces in a more subtle way. It is clear that there is a seemingly endless variety of Gay–Berne homologues, each differing by the values chosen for the four parameters and, in Appendix A, we propose a notation with which to label such potentials. The choice of the ratio κ is relatively straightforward, since for real liquid crystal systems the length-to-breadth ratio of the constituent molecules must be equal to or greater than about 3:1. Accordingly, κ was set equal to 3 by Gay and Berne.⁴ The selection of the remaining three parameters is not so straightforward and to obtain them the potential was compared to that for a pair of particles, each formed by four Lennard–Jones sites placed equally along a line, such that the length-to-breadth ratio was 3:1. This gave κ' as 5 and the exponents μ and ν as 2 and 1, respectively.⁴ There are two problems with this composite Lennard–Jones model; first, the shape of a Gay–Berne particle is ellipsoidal whereas that for the four Lennard–Jones centers approximates to a spherocylinder. Secondly, the molecular structure of a typical nematogen, such as 4,4'-dimethoxyazoxybenzene, is significantly more complex than that of the composite Lennard–Jones model. Despite these problems, the original parametrization of the Gay–Berne potential, or modest variations on the parameters, have proved to be remarkably successful. It has a relatively rich phase behavior with isotropic, nematic and smectic B phases having been clearly identified.^{6–8} In addition, there are claims to have observed a smectic A,⁷ a tilted smectic B⁸ and a rippled smectic B phase.⁹ The properties of the phases and the transitions between them also appear to be in reasonable agreement with those found for real mesogens. Such success has contributed to an increasing use of this potential model and its further development; for example, these developments include the introduction of dipolar forces,¹⁰ flexibility,¹¹ the construction of more complex particle shapes based on a collection of Gay–Berne sites¹² and the extension to biaxial particles.¹³

Despite the apparent success of the original parametrization of the Gay–Berne potential, it is not clear how typical the parameters are in comparison to those obtained by fitting the potential to real rodlike mesogenic molecules rather than the composite Lennard–Jones model. To explore this facet of the model, Luckhurst and Simmonds¹⁴ selected *p*-terphenyl and, although this is not a liquid crystal at atmospheric pressure, it does have a virtual nematic–isotropic transition temperature of 360 K. In addition, the molecule has the advantage of being relatively rigid although, like all mesogenic molecules, it has a slight biaxiality which had to be projected out in order to determine the parameters in the uniaxial Gay–Berne potential.¹⁴ The values so determined are $\kappa=4.4$, $\kappa'=39.6$, $\mu=0.80$, and $\nu=0.74$. The length-to-breadth ratio is significantly larger than that for the original parametrization but the increase in the well depth ratio is even more dramatic. A system of Gay–Berne particles interacting with this new set of parameters was studied at a limited number of state points and found to form an isotropic, a nematic and, quite definitely, a smectic A phase.¹⁴ It was also discovered that reduction of κ' from its high value of 39.6 did not appear to influence the formation of the smectic A phase until it was less than 5.0 when only a nematic phase was

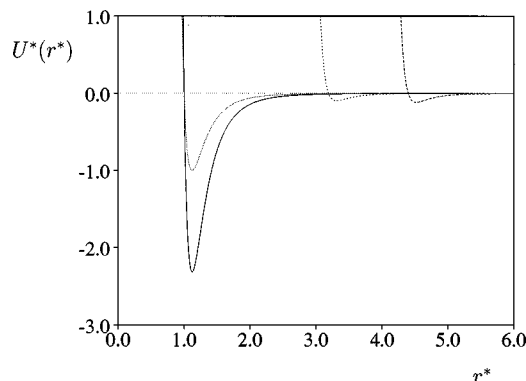


FIG. 1. The dependence of the scaled Gay–Berne potential, $U^*(r^*)$ ($\equiv U(r^*)/\epsilon_0$), on the scaled separation, r^* ($\equiv r/\sigma_0$), for the parameters $\kappa=4.4$, $\kappa'=20.0$, $\mu=1$, $\nu=1$ for a pair of particles in the side-by-side (—), end-to-end (---), cross ($\cdot \cdot \cdot$), and side-to-end (- · - ·) configurations.

formed. Such behavior suggests that the formation of the smectic A phase, in comparison with the smectic B phase of the original parametrization of the Gay–Berne potential, results from the increase in the length-to-breadth ratio. This explanation is supported by subsequent studies of a series of Gay–Berne mesogens in which the ratio κ was changed from 3.0 to 4.0, while keeping the other parameters fixed at the original values suggested by Gay and Berne.¹⁵

In view of the wide occurrence and importance of the smectic A phase for real mesogens,¹⁶ together with the fact that the new parametrization of the Gay–Berne model was only investigated for a small number of state points, we have undertaken a far more detailed study of an analogous mesogen. The phase behavior and the characterization of the phases are reported here. For computational convenience, the exponents μ and ν were both set equal to unity, instead of their nonintegral values. The value of κ was retained at 4.4 and κ' was reduced to 20 which, as we have seen, should not influence the formation of the smectic A phase. The form of the potential function for this choice of parameters is shown in Fig. 1 for four orientations of the particles; side-by-side, end-to-end, cross, and side-to-end. Two features are immediately apparent; one is the strong anisotropy in shape of the Gay–Berne particle, which is indicated by the large variation in the contact distance with the orientations of the particles, and the other is the large difference in the well depth for the side-by-side and the end-to-end configurations. Indeed, it is because the well depth for the end-to-end configuration is so shallow that even if κ' is halved, although this effectively doubles the well depth, the absolute change is so small that it would not be expected to influence the phase behavior significantly. We note that the contact distance and the well depth for the cross configuration give the scaling parameters σ_0 and ϵ_0 (see Fig. 1 and Appendix A).

In presenting the results for this Gay–Berne model mesogen, we begin with the phase behavior and its dependence on pressure; we then consider in detail the structure of these phases. The format of the paper is as follows. The essential details of the isothermal–isobaric Monte Carlo simulations are described in Sec. II, which also discusses the order parameters used to distinguish between the isotropic, the nem-

atic and the smectic phases. The results of the simulations are given in Sec. III, where the transitional properties are compared with experiment and also used to construct the phase diagram as a function of temperature and pressure. The detailed structures of the liquid crystal phases have also been investigated for considerably larger systems. These were studied using molecular dynamics simulations at constant volume and temperature; the details of these simulations are given in Sec. IV. This section also describes the distribution functions and correlation coefficients employed to characterize the phases. The results of the molecular dynamics simulations are given in Sec. V, where they are compared with the predictions of molecular field theories of liquid crystals. The prime results of the simulations are summarized in Sec. VI, which also contains the conclusions of our study of this new parametrization of the Gay–Berne potential.

II. ISOTHERMAL–ISOBARIC MONTE CARLO SIMULATIONS: COMPUTATIONAL DETAILS AND ORDER PARAMETERS

We have used isothermal–isobaric or constant pressure Monte Carlo simulations¹⁷ to investigate the phase behavior of the Gay–Berne (GB) mesogen with parameters $\kappa=4.4$, $\kappa'=20.0$, $\mu=1$ and $\nu=1$. Although constant pressure simulations are slightly more computationally expensive than canonical or constant volume simulations, they do possess a number of advantages. In a constant volume simulation, the natural structure of the phase may not fit the exact dimensions of the simulation box, especially for small systems. This means that the particles may deviate from their preferred positions in order to be commensurate with the periodic images due to the boundary conditions.⁷ This can lead to a deformation of a translationally ordered phase; indeed, this could explain the tilted smectic B phase observed for the GB(3.0,5.0,2,1) mesogen.⁸ Another problem associated with constant volume simulations is that, by definition, the system cannot change density. If a system is restricted to a particular density, then it is often found that cavities appear at low temperatures because the natural crystal structure has a higher density. Conversely, if a higher density is used, then it is possible that the crystal may not melt, even at reasonably high temperatures. Constant pressure simulations overcome both of these problems by allowing the box shape and volume to change. The removal of the constant box shape constraint typically used in constant volume simulations allows, for example, the layers in a smectic phase to achieve their preferred spacing.⁹ A further problem occurs in constant volume simulations for systems exhibiting first order transitions, such as that between the nematic and isotropic phases. Here, the phases coexist over a range of temperatures at constant density and, as a result, experimental observables, such as the internal energy per particle, are found to be continuous in the region of the transition, even though this is first order. As a consequence the transitional region is hard to locate and the properties at the transition are difficult to determine. At constant pressure, however, the transition from one phase to the other is discontinuous, although the behavior observed in the simulation cannot be absolutely discrete because of the finite size of the system. Finally, we note that experiments on

real liquid crystals are usually performed at constant pressure; the isothermal–isobaric ensemble is, therefore, closer to normal experimental conditions than the canonical ensemble, thus facilitating comparison with experiment.

To investigate the phase behavior of the mesogen GB(4.4,20.0,1,1), we have performed simulations on a system of $N=2000$ particles at scaled pressures, P^* ($\equiv P\sigma_0^3/\epsilon_0$), of 1.0, 2.0 and 3.0. The simulations were started from an isotropic configuration, obtained by melting a simple cubic lattice at an appropriate temperature for each pressure. The scaled temperature, T^* ($\equiv k_B T/\epsilon_0$), was then reduced in small steps to explore the range of phases formed. After each temperature change, equilibration runs of typically 50 000 cycles were performed, although this value was by no means fixed; each cycle consisted of N trial single particle moves at constant volume followed by a trial volume and box shape change, in which the dimension of a random box axis was altered randomly. Longer equilibration runs of 100 000–250 000 cycles were found to be necessary near the transitions to an ordered phase, whether this be orientational order (nematic–isotropic) or translational order (smectic A–nematic). At each temperature, the equilibration stage was followed by a production run of either 20 000 cycles in the isotropic phase or 50 000 cycles in the liquid crystal phases. To check for hysteresis in the thermodynamic observables, heating runs were performed at pressures P^* of 2.0 and 3.0. These simulations were started from an ABAB hexagonal crystal structure and heated in small steps. The heating and cooling runs were found to give essentially identical results which provides further evidence that the systems were well equilibrated.

The trial particle moves were generated by a random displacement along each box axis, coupled with a random rotation using the method of Barker and Watts.¹⁸ In the nematic phase, the maximum scaled displacement along each axis was typically 0.08–0.12 and the maximum rotation 6° – 8° . The maximum allowed box change in an equilibrated phase was typically 0.1% of the box length. A potential cutoff, r_c^* ($\equiv r_c/\sigma_0$), of 5.5 and a Verlet neighbor list¹⁹ were used to speed up the calculations. For the side-by-side configuration, the scaled potential energy at this cutoff is -0.00033 , which is just 0.014% of that at the well minimum for this configuration. At the other extreme, for the end-to-end configuration, the scaled potential energy at the cutoff is -0.0053 , which corresponds to 4.6% of that at the well minimum. Although this fraction may appear large, it should be recalled that the energy at the minimum is only -0.1156 (see Fig. 1) and that it is the side-by-side rather than the end-to-end configuration which is expected to dominate the phase behavior. The constant pressure Monte Carlo simulations for 2 000 particles typically took 1.6 s per cycle for the nematic phase at a scaled density, $\langle\rho^*\rangle$ ($\equiv \langle\rho\rangle\sigma_0^3$), of 0.175 on a single processor of a Silicon Graphics Power Challenge.

For the translationally ordered smectic phases, we have found that the equilibration stage is more efficient if the director is aligned along a box edge because the layer spacing is free to change in trial moves independent of the average side-by-side particle separation.⁹ Once an orientationally or

dered phase was identified, the director was aligned by a field applied along the z axis before further cooling; here the potential for each particle was modified by the addition of the term

$$U_f(\hat{\mathbf{u}}_i) = -\lambda_f(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{z}})^2, \quad (1)$$

where $\hat{\mathbf{z}}$ is a unit vector along z . The system was given a further equilibration period of 50 000 cycles after the aligning field had been turned off. In such relatively large equilibrated systems, the rotational motion of the director is extremely slow compared to that of the particles and accordingly the director orientation is found to remain essentially unchanged on further cooling. This procedure can, therefore, be used to align the director without influencing the subsequent thermodynamics of the system. We note that this method is not appropriate for canonical simulations with a constant box shape because the (unknown) periodicity of the layer spacing is unlikely to fit the z dimension of the box.

Two thermodynamic properties, the scaled enthalpy per particle, $\langle H^* \rangle (= \langle H \rangle / \epsilon_0)$, and the scaled volume per particle, $\langle V^* \rangle (= \langle V \rangle / \sigma_0^3)$, were used to characterize the transitions between the phases. In addition, we have calculated two long range order parameters to distinguish between the phases formed and to locate their transitions. One is the second rank orientational order parameter, $\langle P_2 \rangle$, where $P_2(\cos \beta)$ is the second Legendre polynomial and β is the angle between the director and the symmetry axis of the particle.²⁰ This order parameter vanishes in the isotropic phase, is nonzero in the liquid crystal phases and is unity in the limit of complete orientational order. The other quantity is the translational order parameter, $\langle \tau_1 \rangle$, which is equal to the average $\langle \cos(2\pi z/d) \rangle$, where z is the coordinate of a particle along the director and d is the layer spacing.^{20,21} This order parameter vanishes in the isotropic and nematic phases and is nonzero in the smectic phases. Provided the origin is located at the center of a smectic layer and the average is taken over an integral number of layers, $\langle \tau_1 \rangle$ is positive and equal to unity in the limit of complete one dimensional translational order.

The orientational order parameter, $\langle P_2 \rangle$, was calculated via the $\mathbf{Q}$ -tensor introduced by Vieillard-Baron.²² In the isotropic phase, the value of this order parameter is small but nonzero because of the finite system size, as discussed by Eppenga and Frenkel.²³ If $\langle P_2 \rangle$ is calculated from the largest eigenvalue, its value is typically found to be 0.05 for our system of 2000 particles; however, the magnitude is reduced to 0.02 or less when calculated from the middle eigenvalue. In an orientationally ordered phase, both of these eigenvalues yield essentially the same value for the order parameter, which also indicates that the phase has uniaxial symmetry. We have, therefore, used the middle eigenvalue to evaluate $\langle P_2 \rangle$, especially as the error in this quantity is more sensitive to the system size.²³

The calculation of the translational order parameter

$$\langle \tau_1 \rangle = \langle \cos 2\pi z/d \rangle, \quad (2)$$

from the simulation is a nontrivial task especially in the vicinity of the transition to the smectic phase. Indeed it would seem that we need to know the layer spacing, d , even before

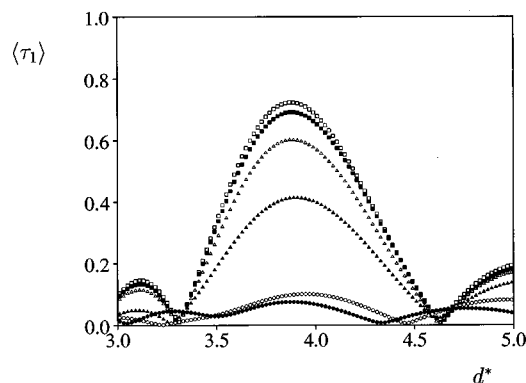


FIG. 2. The dependence of the translational order parameter, $\langle \tau_1 \rangle$ [see Eq. (3)], on the scaled layer spacing, d^* , on either side of the smectic A–nematic transition at scaled temperatures, T^* , of 1.30 ($\square$), 1.35 ($\blacksquare$), 1.40 ($\triangle$), 1.45 ($\blacktriangle$), 1.50 ($\circ$), and 1.55 ($\bullet$).

the order parameter, which can identify the phase as a smectic, has been determined. However, this proves not to be the case. For an orthogonal smectic phase, such as a smectic A, the normal to the layers is parallel to the director, which identifies the coordinate frame in which the order parameter is to be calculated. Then to determine the layer spacing we would need to find a value of d which maximizes $\langle \tau_1 \rangle$ [see Eq. (2)], since this ensures that the periodicity in the function being averaged is identical to that in the translational distribution function. However, this still requires that the origin of the z coordinates is known and this difficulty is compounded by the possibility that the position of the smectic layers within the box can shift during the course of the Monte Carlo simulation, especially for small systems. This problem can be overcome by using the more general definition of the translational order parameter for a one dimensional system; this is²¹

$$\langle \tau_1 \rangle = |\langle \exp(2\pi iz/d) \rangle|, \quad (3)$$

which is independent of the choice of the origin for the z coordinates. Now the value of d can be evaluated by maximizing this more general expression for the translational order parameter; this is achieved numerically because the z coordinate is divided by the layer spacing in the definition of the order parameter. The maximum value of $|\langle \exp(2\pi iz/d) \rangle|$ then gives the translational order parameter.

To illustrate the application of this procedure to determine $\langle \tau_1 \rangle$, we preempt some of the results given in the following section; there we show that, at a scaled pressure of 2.0, the mesogen GB(4.4,20.0,1,1) forms both smectic A and nematic phases. Taking configurations from the production stage of these simulations at temperatures both above and below T_{SmAN}^* , we have calculated $\langle \tau_1 \rangle$ defined by Eq. (3) for scaled layer spacings, $d^* (= d/\sigma_0)$, in the range 3.0–5.0. This includes the length of the particle 4.4, which experience of real smectogens suggests should be close to the layer spacing. The dependence of $\langle \tau_1 \rangle$ on d^* is shown in Fig. 2 for a selection of temperatures on either side of the transition. At T^* of 1.45, the order parameter exhibits a clear maximum at a layer spacing of about 3.9, where $\langle \tau_1 \rangle$ is ~ 0.4 . For lower temperatures the order parameter at the

maximum grows significantly, although there is only a small variation in d^* , which decreases slightly. At T^* of 1.50 and above, there is also a maximum in the order parameter, although this is clearly rather weak ($\langle\tau_1\rangle\sim 0.1$) and at a value of d^* which changes strongly with temperature. We attribute this behavior to the relatively small system size, which means that even in the nematic phase there is still some artificial smecticlike ordering. Such behavior is analogous to the small value of $\langle P_2 \rangle$ found in simulations of the isotropic phase. Nonetheless, it is clear that the method provides a valuable route to the translational order parameter, $\langle\tau_1\rangle$, and, therefore, to the location of the transition between the nematic and smectic A phases; as we shall see, this may be difficult to determine from other quantities, such as the enthalpy, for this weak transition.

Finally, we note that there is another more direct way to determine the translational order parameter since, in practice, the positions of the smectic layers do not change during the Monte Carlo simulations with 2000 particles. This means that it is relatively straightforward to obtain the singlet translational distribution function. The periodicity, d^* , can be determined directly from this and the convolution of the distribution function with the appropriate Chebychev polynomial will give all of the translational order parameters.

III. RESULTS AND DISCUSSION: THERMODYNAMIC PROPERTIES AND ORDER PARAMETERS

The identification of the liquid crystal phases formed by the mesogen GB(4.4,20.0,1,1) is considered in detail in Sec. V. Here we are primarily concerned with the transitions between the phases, although clearly the availability of the order parameters, $\langle P_2 \rangle$ and $\langle\tau_1\rangle$, allows provisional assignment of the phases. At a scaled pressure of 1.0, the system is found to exhibit smectic A and smectic B phases on cooling from the isotropic phase. The temperature dependencies of the enthalpy per particle, $\langle H^* \rangle$, the volume per particle, $\langle V^* \rangle$, and the second rank orientational order parameter, $\langle P_2 \rangle$, are shown in Fig. 3. It is apparent that a transition occurs at a temperature, T^* , of 1.15 ± 0.05 , where the uncertainty is just half of the difference between the two temperatures on either side of the transition. The enthalpy of transition, $\Delta\langle H^* \rangle$, was estimated by extrapolating the results for $\langle H^* \rangle$ in both phases back to the transition. This gives a value of 2.86 ± 0.04 , which corresponds to an entropy change, $\Delta S/R$, of 2.49 ± 0.05 . This is certainly large in comparison to the typical experimental values of 0.2–0.3 observed for the nematic–isotropic transition, but is comparable to those of 1.5–3.0 found for the smectic A–isotropic transition.²⁴ This quantitative difference in the transitional enthalpies between the N–I and the SmA–I transitions is due primarily to the existence of translational order in the smectic A phase. We note that the entropy change, $\Delta S/R$, at the melting transition for real mesogens is typically 8–12, which supports the conclusion that this transition is to a partially ordered smectic rather than a crystal phase. The relative change in volume at the smectic A–isotropic transition is defined as $2(V_I^* - V_{SmA}^*)/(V_I^* + V_{SmA}^*)$; this is found to be 14.8% for the mesogen GB(4.4,20.0,1,1). This is similar to the value of 14.0% found for the smectic B–isotropic transition exhibited

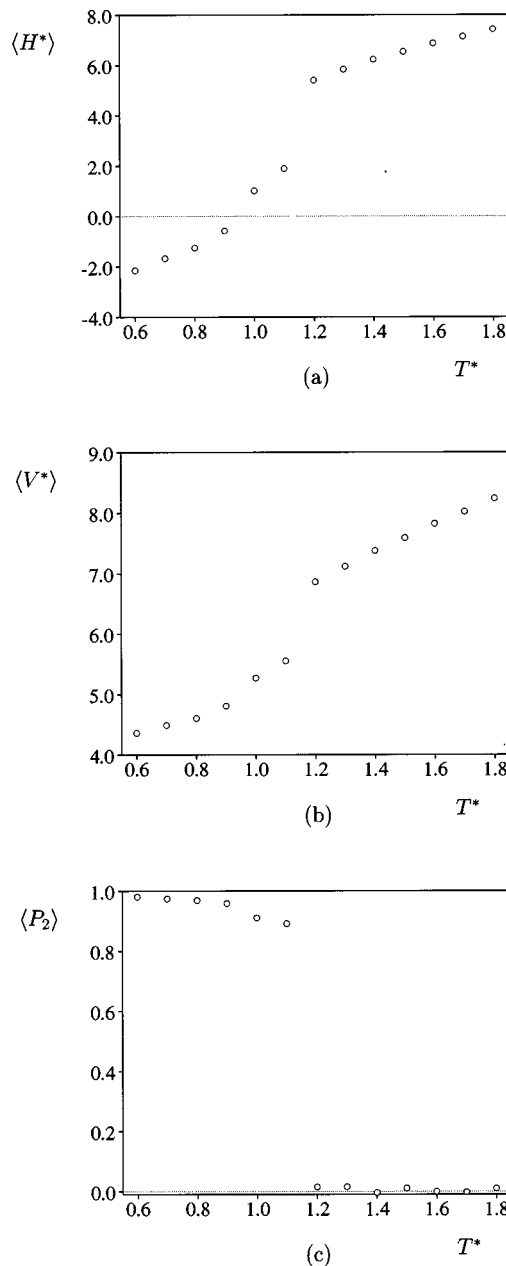


FIG. 3. The temperature dependencies of (a) the scaled enthalpy per particle, $\langle H^* \rangle$, (b) the scaled volume per particle, $\langle V^* \rangle$, and (c) the second rank orientational order parameter, $\langle P_2 \rangle$, at a scaled pressure, P^* , of 1.0.

by the mesogen GB(3.0,5.0,2,1) at a scaled pressure of 1.0.⁹ The transition between the smectic A and smectic B phases, at T^* of 0.95 ± 0.05 , is found to be relatively weak. The scaled enthalpy change at this transition is estimated to be 0.86 ± 0.04 , which leads to a transitional entropy, $\Delta S/R$, of 0.90 ± 0.04 . The relative change in the volume at the SmB–SmA transition is found to be $\sim 5.0\%$, which is consistent with the small transitional entropy. We finish with the second rank orientational order parameter, $\langle P_2 \rangle$. It is apparent from Fig. 3 that the transition at T^* of 1.15 is to a phase with a high orientational order; there is a jump in $\langle P_2 \rangle$ from essentially zero to 0.90, which is in keeping with the large volume and enthalpy change at the SmA–I transition. This is somewhat larger than the order parameter measured by

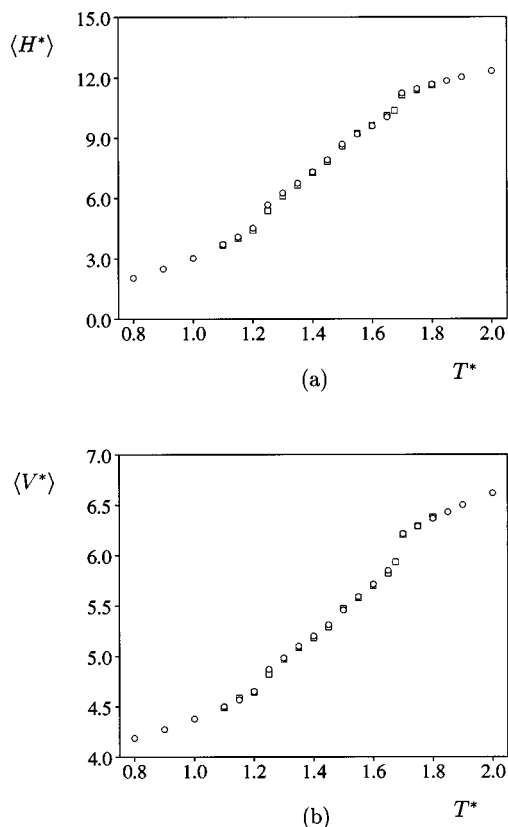


FIG. 4. The temperature dependencies of (a) the scaled enthalpy per particle, $\langle H^* \rangle$, and (b) the scaled volume per particle, $\langle V^* \rangle$, at a scaled pressure, P^* , of 2.0. The results were obtained for both cooling ($\circ$) and heating ($\square$) runs.

Doane *et al.*²⁵ for a long chain 4-*n*-alkoxybenzylidene-4'-phenylazoaniline but is comparable to that of ethyl-4-azoxybenzoate estimated using a nitroxide spin probe.²⁶ However, such differences are expected because, unlike the nematic-isotropic transition, the smectic A-isotropic transition is not predicted by molecular field theories to exhibit universal behavior.

We now turn to the phase behavior at a pressure P^* of 2.0; here, we observe the phase sequence SmB-SmA-N-I. The temperature dependencies of the enthalpy, $\langle H^* \rangle$, and the volume per particle, $\langle V^* \rangle$, are shown in Fig. 4. It is apparent that there are discontinuities in the enthalpy at temperatures, T^* , of 1.225 ± 0.025 (SmB-SmA) and 1.675 ± 0.025 (N-I), although, as we shall see, there is also a much weaker transition at a temperature, T^* , of 1.475 ± 0.025 (SmA-N). The enthalpy change, $\Delta\langle H^* \rangle$, at the nematic-isotropic transition is estimated to be 0.80 ± 0.02 . The corresponding transitional entropy, $\Delta S/R$, of 0.47 ± 0.02 is characteristic of a transition between the nematic and the isotropic phases. This is about double the value of 0.20 found for 4,4'-dimethoxyazoxybenzene at atmospheric pressure;²⁷ it is, however, in good agreement with the prediction of 0.417 made by the Maier-Saupe theory.²⁸ The strength of the transition between the two smectic phases is of comparable magnitude; we estimate $\Delta\langle H^* \rangle$ and $\Delta S/R$ to be 0.68 ± 0.02 and 0.55 ± 0.02 , respectively. The enthalpy change between the nematic and smectic A phases is much

harder to measure because of the weak nature of this transition; we estimate it to be 0.24 ± 0.10 , which gives a transitional entropy, $\Delta S/R$, of 0.16 ± 0.08 . According to theory, the strength of the smectic A-nematic transition is determined by the ratio of the transition temperatures, T_{SmA-N}/T_{N-I} , which reflects the extent of the orientational order in the nematic phase just prior to the transition into the smectic A phase.²⁹ For the mesogen GB(4.4,20.0,1,1), the ratio of transition temperatures is 0.87; studies of the 4-*n*-alkoxybenzylidene-4'-phenylazoanilines²⁵ have shown that the smectic A-nematic transition is extremely weak for such a wide nematic range in accord with the simulation results. The simplest version of the McMillan theory of the smectic A phase²⁹ predicts that, for a ratio T_{SmA-N}/T_{N-I} of less than or equal to 0.89, the transition should be second order, which is close but not quite in agreement with our results. To achieve agreement with these, it would be necessary to use an extension of the McMillan theory to include a pure translational term, which then introduces an adjustable parameter.³⁰ This has the effect of either raising or lowering the tricritical temperature at which the smectic A-nematic transition becomes second order. The relative volume changes at the N-I, SmA-N, and SmB-SmA transitions are estimated to be 4.2%, 0.8%, and 2.5%, respectively, and seem to be consistent with the entropy changes at these transitions. We note that the relative volume change between the nematic and the isotropic phases is rather large compared with those of real systems; indeed for 4,4'-dimethoxyazoxybenzene, this is one order of magnitude smaller at 0.3%.²⁷ This significant quantitative failure of the Gay-Berne model is somewhat difficult to understand but it may be associated with the ellipsoidal shape of the particles rather than the anisotropy in the repulsive forces.

The second rank orientational order parameter, $\langle P_2 \rangle$, shown in Fig. 5(a), exhibits a jump from zero to ~ 0.45 at the nematic-isotropic transition. Unlike the volume change, the order parameter is close to that found for real nematogens. For example, for 4,4'-dimethoxyazoxybenzene, the second rank orientational order parameter at the nematic-isotropic transition has been found to be 0.40 and independent of pressure up to 3000 atm,³¹ although other studies indicate that the transitional order parameter may be somewhat lower at 0.35.²⁷ In addition, the prediction of the Maier-Saupe theory for the order parameter at the transition is 0.429 ,²⁸ which is in good agreement with our result. On further cooling, $\langle P_2 \rangle$ is observed to increase gradually to almost 0.80 within the nematic region; we note that this is slightly higher than that expected for 4,4'-dimethoxyazoxybenzene at the same reduced temperature, T/T_{N-I} .³² However, this is typical of the values observed for other model mesogens.^{6-8,14}

As we have seen, the smectic A-nematic transition is extremely weak and so we require an order parameter which can distinguish unambiguously between these phases, so as to locate the transition. The translational order parameter, $\langle \tau_1 \rangle$, [see Eq. (2)] satisfies this requirement, since it is zero in the nematic phase and is unity for a perfectly layered smectic A phase. The temperature dependence of $\langle \tau_1 \rangle$ at P^* of 2.0 is shown in Fig. 5(b) (see also Fig. 2). In the isotropic and nematic phases, the order parameter is found to be about

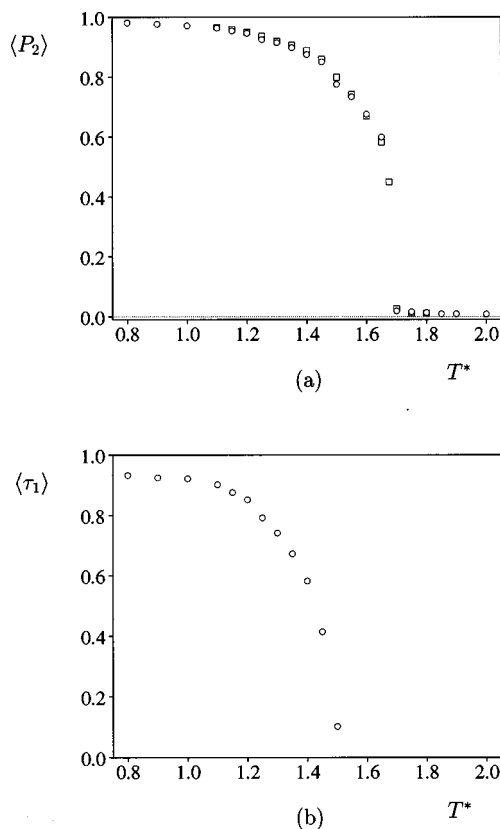


FIG. 5. The temperature dependencies of (a) the second rank orientational order parameter, $\langle P_2 \rangle$, and (b) the translational order parameter, $\langle \tau_1 \rangle$, at a scaled pressure, P^* , of 2.0. The orientational order parameter was determined from both cooling ($\circ$) and heating ($\square$) runs.

0.1 or smaller; this nonzero value is taken to result from the small system size and the fact that the layer spacing is not well defined for these phases. However, at a slightly lower scaled temperature of 1.45, the order parameter adopts the value of 0.40, indicating that there has been a transition to a smectic phase with moderate translational order. This behavior of $\langle \tau_1 \rangle$ is also consistent with our previous determination of the smectic A–nematic transition temperature of 1.475. As the temperature is lowered, so the translational order parameter grows, relatively rapidly, until it reaches a value of about 0.80 just prior to the transition into the smectic B phase. After this, $\langle \tau_1 \rangle$ still increases with decreasing temperature but far less rapidly until, at the lowest temperature we have studied, it is ~ 0.95 , corresponding to a phase with an essentially perfect layer structure. However, we note that this order parameter cannot be used to locate the transition between the smectic phases. In principle, this could be achieved from measurements of the long range limit of the bond orientation correlation function, which we shall discuss in Secs. IV and V.

On increasing the pressure P^* to 3.0, we again observe the formation of isotropic, nematic, and smectic A phases for the mesogen GB(4.4,20.0,1,1). Presumably, a smectic B is also present below the smectic A phase; however, this was not investigated because the system is rather dense in this region of the phase diagram and, therefore, extremely long runs are required to ensure full equilibration. The tempera-

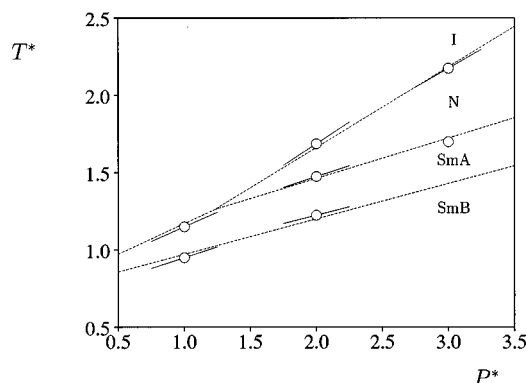


FIG. 6. A sketch of the phase diagram for the mesogen GB(4.4,20.0,1,1); the points show the transition temperatures obtained from the constant pressure Monte Carlo simulations and the solid lines show the phase boundaries calculated from the Clapeyron equation [see Eq. (4)]. The dashed lines indicate the approximate form of the phase diagram.

ture dependencies of the enthalpy, the volume per particle and the order parameters are analogous to those found for the lower scaled pressure of 2.0. The nematic–isotropic transition temperature, T_{NI}^* , is found to be 2.175 ± 0.025 but, although the transition temperature has increased significantly on increasing the pressure, the transitional properties $\Delta S/R$, $2(V_I^* - V_N^*)/(V_I^* + V_N^*)$ and $\langle P_2 \rangle^{NI}$ are essentially unchanged at 0.44, 3.6%, and 0.38, respectively. This insensitivity of the transitional properties to pressure is consistent with experimental studies³¹ and with the predictions of the Maier–Saupe theory.²⁸ The smectic A–nematic transition temperature is also found to increase with the increase in pressure to a value of 1.70 ± 0.10 . The larger uncertainty in determining the transition temperature results from the essentially continuous nature of the transition; this is consistent with the lower value of the ratio of transition temperatures, $T_{SmA/N}^*/T_{NI}^*$, which is 0.795. This corresponds to a longer nematic range and so a higher orientational order before the transition to the smectic phase occurs. In keeping with the prediction of the McMillan theory,²⁹ this requires that the transition is second order.

The results for the pressure dependence of the transition temperatures can be used to construct the phase diagram for this liquid crystal model. The number of data points is somewhat limited but we can use the Clapeyron equation

$$\frac{dP^*}{dT^*} = \frac{\Delta \langle H^* \rangle}{T^* \Delta \langle V^* \rangle}, \quad (4)$$

to estimate the slopes of the phase boundaries; clearly the reliability of such estimates will increase with the strength of the phase transition. The transition temperatures are shown as a function of pressure in Fig. 6, where the predicted slopes of the phase boundaries at these points are drawn as solid lines. These results were then interpolated to give the phase diagram over a range of pressures; however, we should emphasize that this is only expected to be correct at a semiquantitative level. We observe that, at low pressures, the nematic phase is not stable; however, on increasing the scaled pressure to ~ 1.25 , the nematic enters the phase diagram, giving a smectic A–nematic–isotropic triple point. We find that

the slope of the nematic–isotropic boundary is about twice that for the smectic A–nematic boundary and so the temperature range over which the nematic phase is stable increases as the pressure is increased. In contrast, the slopes of the SmB–SmA and the SmA–N phase boundaries are found to be similar, which gives a more or less constant smectic A range.

It is of interest to estimate what the values of the pressures used in our simulations would correspond to for a real liquid crystal. To do this, we require the scaling parameters, σ_0 and ϵ_0 , which occur in the definition of the scaled pressure, $P^*(\equiv P\sigma_0^3/\epsilon_0)$. The distance scaling parameter, σ_0 , can be identified as the width of a molecule; from x-ray diffraction studies,³³ this value is typically 4.5 Å. To estimate the energy scaling parameter, ϵ_0 , we compare the transition temperature from the simulation at a scaled pressure of 2.0 with that for 4,4'-dimethoxyazoxybenzene, which is 409 K at 1 atm.²⁷ This gives a value for $N_A\epsilon_0$ of 2.01 kJmol⁻¹ and so our estimate of the pressure corresponds to ~730 atm. To make a proper comparison, it is necessary to recalculate the transition temperature for the nematogen at this higher pressure and, accordingly, recalculate ϵ_0 ;⁹ we require, therefore, the value of dT_{NI}/dP , which for 4,4'-dimethoxyazoxybenzene is 4.6×10^{-7} K m² N⁻¹.²⁷ This scheme was repeated following the method of Hashim *et al.*,⁹ until convergence was reached. The resulting scaled pressure of 2.0 turns out to be 790 atm and, while large, this is not unrealistic. Our converged estimate for $N_A\epsilon_0$ of 2.19 kJmol⁻¹ leads to a corresponding temperature T_{NI} of 445 K at this pressure. McColl³⁴ has studied this nematogen over a range of pressures up to 2000 atm and, at 443 K, found that the nematic phase was stable over pressures of 800–1200 atm; thus our results do seem to be relatively realistic. However, unlike the mesogen GB(4.4,20.0,1,1), the real nematogen 4,4'-dimethoxyazoxybenzene does not form a smectic phase and the nematic phase remains stable at low pressures. In contrast, the mesogen bis(4,4'-*n*-heptyloxybenzylidene)-1,4-phenylenediamine does exhibit smectic phases as well as a nematic phase and this phase behavior has been studied as a function of pressure up to about 2500 atm.³⁵ The smectic phases include a smectic C, which is a tilted analogue of the smectic A, and a smectic B phase. On increasing the pressure the nematic range is found to increase, while that of the smectic C is essentially constant, which is comparable to the behavior of the mesogen GB(4.4,20.0,1,1) in our simulations. In addition, the greater slope of the nematic–isotropic phase boundary in comparison with that for the smectic C–nematic boundary does suggest that for structurally related mesogens, the nematic phase may vanish at low pressures, again in keeping with our results.

IV. CANONICAL MOLECULAR DYNAMICS SIMULATIONS: COMPUTATIONAL DETAILS AND STRUCTURAL PROPERTIES

To investigate the structural properties of the phases identified in the previous section, we have performed simulations on a significantly larger system of 16 000 particles, at four temperatures taken from the constant pressure simula-

tions at P^* of 2.0. To study these systems, we have used the parallel molecular dynamics program GBMEGA,³⁶ which was run on the IBM SP2 at the University of Southampton. The use of this larger system enables the calculation of the pair distribution functions for large separations; this is especially important for the translationally ordered smectic phases. The molecular dynamics simulations were performed in the canonical (constant NVT) ensemble; the temperature was held constant by rescaling the velocities at each timestep.³⁷ The initial configuration for each of the four temperatures studied was constructed from an equilibrated configuration taken from the corresponding 2000 particle Monte Carlo simulation. These were replicated into new boxes of dimensions $2l_x^* \times 2l_y^* \times 2l_z^*$, where l_x^* , l_y^* , and l_z^* were the dimensions of the original box. As a rough guide to the system size, the smectic configurations contained 12 layers, each of ~1330 particles. The initial configurations were equilibrated for 25 000 time steps to remove the correlations present from replicating the smaller system. The equilibration period was followed by a production run of 100 000 time steps, during which configurations were saved every 200 steps for later analysis. The time step, $\delta t^*(\equiv \delta t(\epsilon_0/m\sigma_0^2)^{1/2})$, used to integrate the equations of motion was taken to be 0.0025 and the moment of inertia, $I_\perp^*(\equiv I_\perp/m\sigma_0^2)$, of the particles was set equal to 1.0, which is consistent with the κ value of 4.4.¹⁴ During the production runs, the pressure was found not to be constant but distributed about P^* of 2.0, as we expect, because the simulations were performed at constant volume rather than constant pressure. However, this pressure distribution was found to be narrow and the average values of the potential energy per particle and the second rank orientational order parameter were similar to those obtained from the constant pressure simulations. We can, therefore, be confident that, although we have used the canonical rather than the isothermal–isobaric ensemble, the state points investigated are essentially equivalent to those studied in our Monte Carlo simulations. The simulation times, in seconds per time step, on eight processors of the IBM SP2 were as follows; SmB ($T^*=1.20$, $\rho^*=0.2110$) 1.90, SmA ($T^*=1.40$, $\rho^*=0.1932$) 1.75, N ($T^*=1.60$, $\rho^*=0.1756$) 1.50 and I ($T^*=1.8$, $\rho^*=0.1562$) 1.35.

The structures of the four phases were characterized using a selection of single particle and pair distribution functions. At the single particle level, we have determined the singlet orientational distribution function, $f(\beta)$, which gives the probability of finding the symmetry axis of a particle at an angle β to the director.²⁰ In the smectic A phase, $f(\beta)$ gives this probability independent of the position of the particle within the smectic layer. The translational distribution function, $\rho(z)$, was also evaluated; this gives the probability of finding a particle with a coordinate z projected along the director in a volume element δz , irrespective of its orientation. Associated with this distribution function is the set of translational order parameters²¹

$$\langle \tau_n \rangle = \langle \cos(2\pi n z/d) \rangle, \quad (5)$$

which is the generalization of the order parameter $\langle \tau_1 \rangle$ defined in Eq. (2). In principle, the positions and the orientations may be correlated and it is possible to explore this with

the singlet distribution function $P(z, \beta)$, which gives the probability of finding a particle at a position z along the director and at an angle β to it.^{20,21} Associated with this mixed distribution function are the mixed orientational-translational order parameters

$$\sigma_{L,n} = \langle P_L(\cos \beta) \cos(2\pi n z/d) \rangle. \quad (6)$$

The symmetry of the phases confines L to even values but there is no such constraint on n ; these mixed order parameters were calculated from the simulation data for L between 0 and 8 with n between 0 and 4. The calculation of these order parameters for the smectic A and smectic B phases is relatively straightforward because neither the director orientation nor the location of the layers in the simulation box were found to change during the simulation. The origin of the z coordinates is the center of a smectic layer and this was located as one of the maxima in $\rho(z)$. This distribution could also have been used to determine the smectic periodicity, but in fact we have calculated this from the dimension of the simulation box along the z axis.

The general pair distribution function needed to describe the structure of a liquid crystal phase composed of rodlike particles is clearly a complex quantity both to calculate and represent graphically. We have, therefore, determined a range of somewhat simpler pair distribution functions and correlation coefficients related to the total distribution. The first of these is the radial distribution function, $g(r)$, which gives the probability that one particle is separated by a distance r from another at the origin, irrespective of their orientation and that of the interparticle vector with respect to the director. This is normalized so that in the absence of correlations, $g(r)$ tends to unity. There are two distribution functions analogous to $g(r)$; one is the longitudinal distribution function, $g_{\parallel}(r_{\parallel})$, which gives the probability of finding a particle at a resolved distance of $r_{\parallel}$ along the director from another at the origin, again independent of the orientations of the two particles and the interparticle vector. This resolved distribution is clearly of value in exploring the layered structure of a phase. Similarly, the structure within the smectic layers can be probed with the in-plane distribution function, $g_{\perp}(r_{\perp})$, where $r_{\perp}$ is the separation between the particles resolved onto a plane orthogonal to the director. The in-plane pair distribution function is defined for a single smectic layer; in the simulation these were identified from the maxima in the singlet translational distribution function and $g_{\perp}(r_{\perp})$ was then averaged over the layers.

We have also calculated the correlation function $g_6(r_{\perp})$, which provides a measure of the bond orientational order in a smectic layer and so allows us to distinguish between a smectic A and a smectic B phase.³⁸ This is defined by

$$g_6(r_{\perp}) = \langle \psi_6(\mathbf{r}_i) \psi_6^*(\mathbf{r}_j) \rangle, \quad (7)$$

where the local bond orientational order for particle i at a position $\mathbf{r}_i$ is given by

$$\psi_6(\mathbf{r}_i) = \frac{\sum_k w(r_{ik}) \exp(6i\theta_{ik})}{\sum_k w(r_{ik})}. \quad (8)$$

In this expression, the summation is over the neighboring particles k of particle i and θ_{ik} is the angle between the

vector $(\mathbf{r}_i - \mathbf{r}_k)$ projected onto the plane normal to the director and a fixed reference axis. In general, it is difficult to identify the six nearest neighbors which are usually required in the calculation of $\psi_6(\mathbf{r}_i)$ for a hexatic phase because, if the local translational order is not perfect, then there may be five or seven nearest neighbors; for a smectic A phase, this number may be as low as four or five depending on our nearest neighbor criterion. To overcome this difficulty in selecting the nearest neighbors, a cutoff function $w(r_{ik})$ is introduced;³⁸ we have used a function which is unity for a separation r_{ik}^* below 1.4 and zero for r_{ik}^* above 1.8, with a linear interpolation in between these two extremes. We have explored analogous weighting functions but have found that the behavior of $g_6(r_{\perp})$ is rather insensitive to the choice of these. The bond orientational correlation function is unity for clusters of particles centered at $\mathbf{r}_i$ and $\mathbf{r}_j$ if the orientations of the bonds in the two clusters are the same. If, however, the bond orientations are uncorrelated then $g_6(r_{\perp})$ will vanish. The bond orientational correlations are short range for a smectic A phase and long range for a smectic B phase. Accordingly, we shall be able to use the long range limit for $g_6(r_{\perp})$ to distinguish between these phases.

The distribution of the interparticle vector with respect to the director provides another characteristic feature of the structure within a liquid crystal phase. It is also an important quantity in molecular field theories of liquid crystals, which assume that the interparticle vector is uniformly distributed.²⁸ To probe this distribution function, we have calculated the distance dependent second rank order parameter³⁹

$$g_2^+(r) = \langle P_2^+(\cos \beta_r) \rangle_r, \quad (9)$$

where β_r is the angle between the director, $\hat{\mathbf{n}}$, and the interparticle vector, $\mathbf{r}$.

The orientational correlations between pairs of particles and the interparticle vector have been investigated using the distance dependent correlation coefficients $g_{L_i L_j m}(r^*)$,⁴⁰ which are defined by

$$g_{L_i L_j m}(r) = 4\pi g(r) \langle Y_{L_i m}^*(\omega_i) Y_{L_j -m}^*(\omega_j) \rangle_r. \quad (10)$$

Here, $Y_{L_i m}^*(\omega_i)$ is a spherical harmonic and ω_i denotes the spherical polar angles made by the symmetry axis of the particle in a frame containing the interparticle vector as the z axis. The choice of the x and y axes is arbitrary because the function being averaged depends on the difference in the azimuthal angles for the two particles. These correlation coefficients adopt particularly simple forms in the limit of large separations when the orientational correlations are lost. This can best be seen by first transforming from the interparticle frame to the laboratory frame

$$g_{L_i L_j m}(r) = 4\pi g(r) \langle D_{mn_i}^{L_i}(\Omega) D_{-m-n_j}^{L_j}(\Omega) \times Y_{L_i n_i}^*(\omega_i) Y_{L_j -n_j}^*(\omega_j) \rangle_r. \quad (11)$$

In this expression, $D_{mn_i}^{L_i}(\Omega)$ is a Wigner rotation matrix, Ω denotes the Euler angles relating the laboratory and interparticle frames and the prime denotes the spherical polar angles made by the symmetry axis of the particle in the laboratory

frame. At separations which are sufficiently large so that the orientations are uncorrelated, the two spherical harmonics can be averaged independently of each other and of the product of the Wigner rotation matrices. When the director is taken to define the z axis of the laboratory frame, the separate averaging gives

$$\lim_{r \rightarrow \infty} g_{L_i L_j m}(r) = (-)^m g(r) \delta_{L_i L_j} \langle P_{L_i} \rangle \langle P_{L_j} \rangle. \quad (12)$$

In the isotropic phase, where the order parameters vanish, the coefficients $g_{L_i L_j m}(r)$ tend to zero for large separations. This is also true in the nematic phase when the ranks L_i and L_j are different. When they are the same, $g_{LLm}(r)$ tends to a limiting value of $(-)^m \langle P_L \rangle^2$, where $g(r)$ has been replaced by its long range limit of unity. This provides another route to the long range orientational order parameters, which avoids the problem associated with the movement of the director during the simulation.

V. RESULTS AND DISCUSSION: STRUCTURE OF THE PHASES

In principle, the structure of a macroscopically anisotropic phase can be described in terms of single particle and pair distribution functions together with the pair correlation coefficients. Before we consider these for the four phases which we have simulated, we note that there is an alternative and possibly more direct route by which a phase might be identified. This involves constructing an image from a configuration taken from the simulation which shows the positions and orientations of the constituent particles. Provided the configuration selected is typical for the phase, the image constructed from it will allow the phase type, defined in terms of the molecular organization, to be identified. For example, using computer graphics techniques, the orientationally ordered nematic phase can be distinguished from the disordered isotropic phase. Similarly, the one dimensional translational order in a smectic phase can be identified. Pictures of configurations taken from the production stage of the molecular dynamics simulations are shown for the four temperatures in Fig. 7, where the particles are depicted as ellipsoids. At a scaled temperature of 1.8, the mesogen GB(4.4,20.0,1,1) forms an isotropic phase, which is plainly evident from the configuration shown in Fig. 7(a). The system is devoid of any long range order, either positional or orientational, although there is short range order, in that there are small clusters containing several particles with their symmetry axes parallel. Within the nematic phase [see Fig. 7(b)], the long range orientational order is clearly apparent, with the particles tending to be parallel to the director although there are certainly a few orthogonal to the director. There is no long range translational order although small clusters of side-by-side particles are apparent. At the lower scaled temperature of 1.4, the one dimensional translational order of an orthogonal smectic phase is clearly apparent, with the director parallel to the layer normal [see Fig. 7(c)]. The imperfect ordering of the particles within the layers is evident although there do not appear to be any particles between the layers. It is also clear that the layers are interdigitated to a certain extent, which is consistent with the layer spacing determined

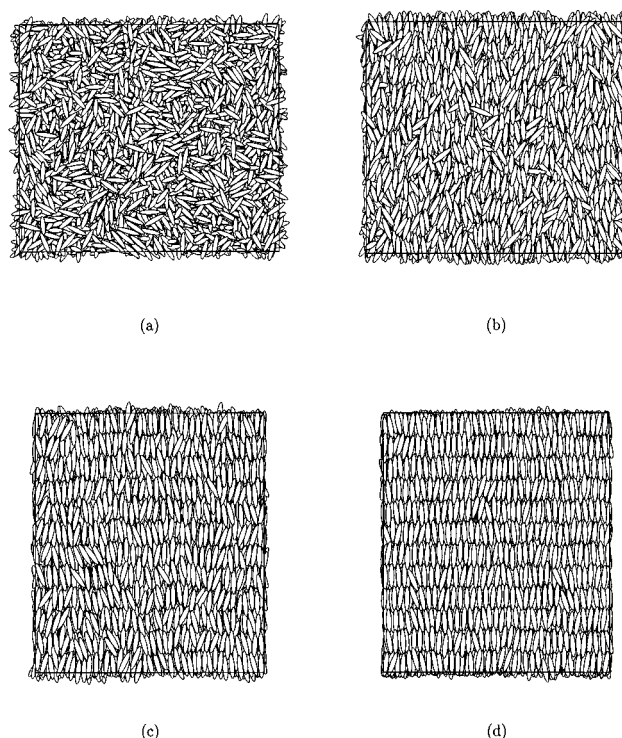


FIG. 7. Pictures of configurations taken from the molecular dynamics simulations of the mesogen GB(4.4,20.0,1,1); (a) the isotropic phase ($T^* = 1.80$, $\rho^* = 0.1562$), (b) the nematic phase ($T^* = 1.60$, $\rho^* = 0.1756$), (c) the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$) and (d) the smectic B phase ($T^* = 1.20$, $\rho^* = 0.2110$), all of which correspond to a scaled pressure, P^* , of ~ 2.0 .

in Sec. III for the smaller system. To identify this phase as a smectic A, it would be necessary to examine the distribution of the particles within a single layer; we observe an essentially disordered liquid layer, although locally the particles do tend to pack in small hexagonal clusters. In the smectic B phase the layer structure is manifestly apparent as is the higher translational and orientational order of the phase [see Fig. 7(d)]. The layers are still interdigitated, as for the smectic A phase; this is presumably aided by the ellipsoidal shape of the Gay-Berne particles. To identify the phase visually as a smectic B, it would again be necessary to view the system along the director to see the distribution of the particles within a single layer. When this is done, the hexagonal distribution of the centers of mass of the particles is clearly apparent.

After this qualitative but nonetheless very informative view of the structure of the four phases exhibited by the mesogen GB(4.4,20.0,1,1), we now turn to a quantitative approach and begin with the singlet orientational distribution function, which provides a complete description of the orientational order at the single particle level. In the isotropic phase, this function is found to be independent of the particle orientation, as expected. This behavior contrasts with that in the liquid crystal phases, where the distribution function is peaked at $\cos \beta$ of one, corresponding to β of zero, as we can see from the results shown in Fig. 8. The width of the distribution function decreases on passing from the nematic, to the smectic A and smectic B phases, in accord with the increase in the orientational order (see Table I); this is also in

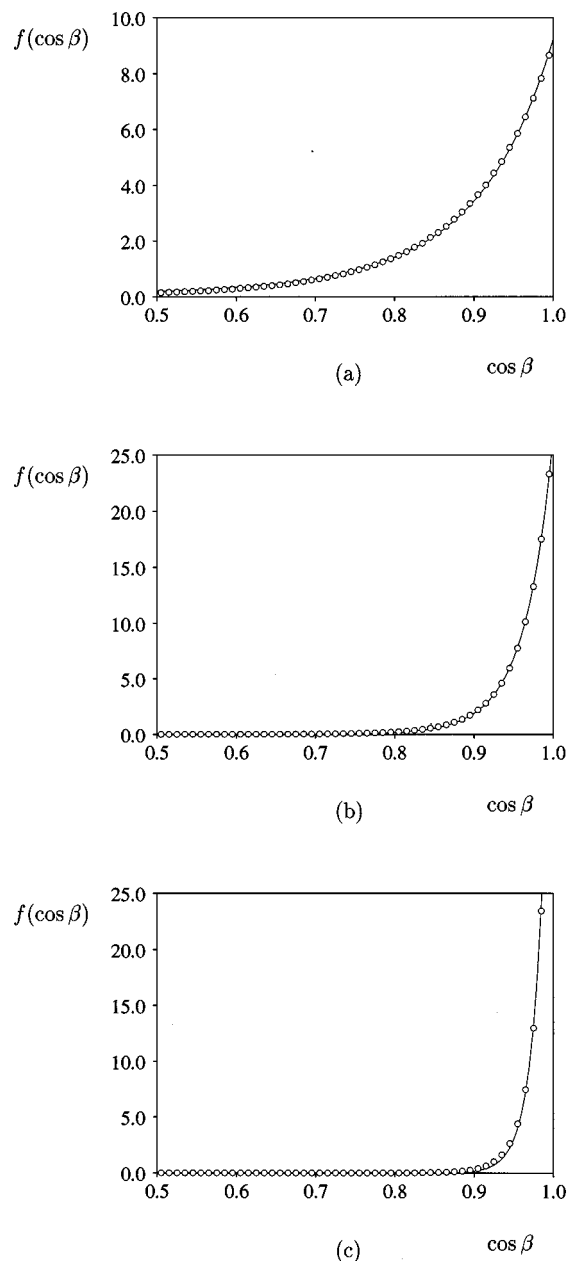


FIG. 8. The singlet orientational distribution function, $f(\cos \beta)$, in (a) the nematic phase ($T^*=1.60$, $\rho^*=0.1756$), (b) the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and (c) the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$), all of which correspond to a scaled pressure of ~ 2.0 . Note that the $\cos \beta$ axis extends from 1.0 to 0.5 rather than 0.0, since the distribution function is strongly peaked about 1.0. The simulation data are shown as points and the best fit Maier-Saupe distribution [see Eq. (15)] as the solid line.

TABLE I. The orientational order parameters, $\langle P_L \rangle$, for the nematic phase ($T^*=1.60$, $\rho^*=0.1756$), the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$) of the mesogen GB(4.4, 20.0, 1, 1).

	$\langle P_2 \rangle$	$\langle P_4 \rangle$	$\langle P_6 \rangle$	$\langle P_8 \rangle$
N	0.689	0.302	0.104	0.028
SmA	0.884	0.668	0.437	0.252
SmB	0.946	0.832	0.682	0.524

agreement with the pictures showing the organization of the particles in these phases (see Fig. 7).

There are various theoretical accounts of the singlet orientational distribution function for uniaxial phases such as the nematic. One of the most general of these is based on the variational principle.⁴¹ In this approach, the Helmholtz free energy is constructed from the unknown singlet orientational distribution function; at the molecular field level, this gives the entropic contribution as

$$S = -k_B \int f(\beta) \ln f(\beta) \sin \beta d\beta. \quad (13)$$

The energetic contribution is assumed to be determined by the orientational order. For a nematic phase, especially near the transition into the isotropic phase, the dominant order parameter is $\langle P_2 \rangle$ and so the energetic contribution is taken to be

$$U = -\frac{1}{2} u_2 \langle P_2 \rangle^2. \quad (14)$$

Minimization of the free energy, constructed from these two contributions, with respect to $f(\beta)$, subject to the constraint that this is normalized, gives

$$f(\beta) = Z^{-1} \exp(a_2 P_2(\cos \beta)), \quad (15)$$

where a_2 depends on the orientational order parameter, $\langle P_2 \rangle$, and Z is the normalization factor. An equivalent result is also given by the Maier-Saupe molecular field theory of nematics,²⁸ although here the form predicted for $f(\beta)$ results from the assumption that the anisotropic intermolecular potential is dominated by second rank terms. We have fitted the distribution function given in Eq. (15) to our data for the nematic phase and the best fit is shown in Fig. 8(a). The agreement between the simulation results and the simple theory is essentially perfect, as has also been found for the mesogen GB(3.0,5.0,2,1).³⁹ The success of the theoretical form of the distribution in accounting for our results for the nematic suggests that it might be equally successful for the smectic A and smectic B phases. The best fits for these two phases are shown in Figs. 8(b) and 8(c) and the agreement is again seen to be good. This result is, however, somewhat surprising because, for such highly ordered systems, although the largest order parameter, $\langle P_2 \rangle$ is certainly not dominant (see Table I). Accordingly, the variational analysis with the additional order parameters contributing to the internal energy [see Eq. (14)], would lead to

$$f(\beta) = Z^{-1} \exp\left(\sum_{L \text{ even}} a_L P_L(\cos \beta)\right). \quad (16)$$

Use of this extended distribution, with the term in $P_4(\cos \beta)$ as well as in $P_2(\cos \beta)$, certainly improves the agreement with the simulated distribution, but the change is found to be slight. We attribute this behavior to the high orientational order of the smectic phases, which means that $f(\beta)$ only adopts significant values for β close to zero. This rapid decay in the singlet orientational distribution function ensures that the terms in the sum of Legendre polynomials only contribute when β is small; each polynomial can, therefore, be expanded and the series truncated after the first nontrivial term. This gives $f(\beta)$, in the high order limit, as⁴²

$$f(\beta) = Z^{-1} \exp(-\lambda_2 \beta^2), \quad (17)$$

where λ_2 is given by

$$\lambda_2 = \sum_{L \text{ even}} \sum_{p=0}^{1/2L} (\tfrac{1}{2}L - p) a_L b_{Lp} \quad (18)$$

and

$$b_{Lp} = \frac{(-)^p (2L - 2p)!}{2^L p! (L - p)! (L - 2p)!}. \quad (19)$$

However, as any Legendre polynomial contributes a term in β^2 , the form for $f(\beta)$ obtained from the complete expansion [see Eq. (16)] is identical to that found for a single $P_L(\cos \beta)$ and so for that with only $L=2$, which we have used. We expect, therefore, that in the high order limit the distribution function given in Eq. (15) will give a good fit to the simulated distribution, even though higher rank Legendre polynomials could be included in $f(\beta)$ [see Eq. (16)]. For those not familiar with the Legendre polynomials we give their forms for some of the even members in Appendix B.

The defining quantity for the smectic phases is their layer structure, in other words, their one dimensional translational order. At the single particle level, the most complete description of this is provided by the one dimensional singlet translational distribution function, $\rho(z)$. Our results for this distribution are shown in Fig. 9. The lack of translational order in the nematic phase is clearly apparent. The distribution in the smectic A phase is rather broad but becomes significantly sharper in the smectic B phase. Such a change is to be expected and is clearly evident from the pictures of configurations taken from the simulation (see Fig. 7). For the smectic A phase, there is a small but finite probability that a particle can be found between the layers. In contrast, there is a negligible probability of finding a particle between the layers in the smectic B phase.

Before we consider the theoretical prediction of the singlet translational distribution function, we first recall the formal relationship between $\rho(z)$ and the translational order parameters, $\langle \tau_n \rangle$, defined by Eq. (5). This is given by the Fourier series²¹

$$\rho(z) = d^{-1} + 2d^{-1} \sum_{n \neq 0} \langle \tau_n \rangle \cos(2\pi n z/d), \quad (20)$$

the convergence of this series is clearly determined by the magnitude of the order parameters. For the smectic A phases formed by real mesogens, it is usual to see only the first order reflection in the x-ray scattering pattern.³³ However, it is important to recognize that the intensity of a particular order of reflection depends upon the molecular structure factor associated with that order as well as the relevant translational order parameter.⁴³ In general, the structure factors decrease with the order of reflection and so the dominance of the first order reflection does not necessarily mean that the translational distribution function is essentially sinusoidal, that is, retention of just the first nontrivial term ($n=1$) in the expansion need not necessarily provide a good description of $\rho(z)$. The translational order parameters for both smectic phases are given in Table II and, using the values of $\langle \tau_1 \rangle$ given here, we have calculated $\rho^*(z^*)$. The results of these

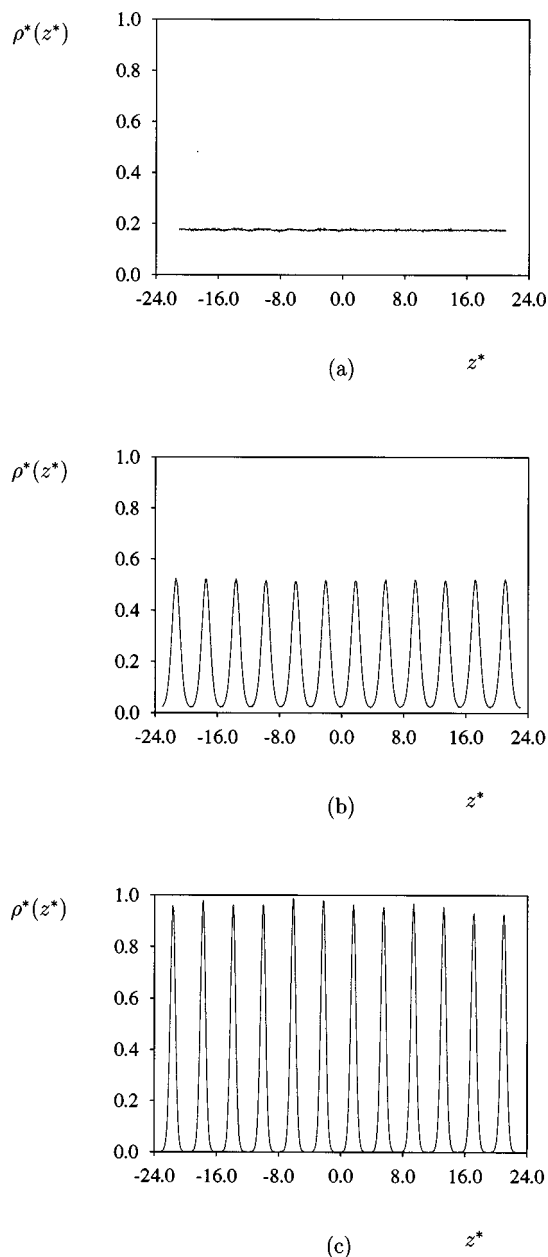


FIG. 9. The one dimensional singlet translational distribution function, $\rho^*(z^*)$, for (a) the nematic phase ($T^*=1.60$, $\rho^*=0.1756$), (b) the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and (c) the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$).

calculations are shown in Fig. 10(a), where $\rho^*(z^*)$ is plotted as a function of z^*/d^* in the range 0 to 1, from the center of one layer to that of the next, averaged over all layers. We see that, for the smectic A phase, the truncated distribution function provides a reasonable but far from complete representation of $\rho^*(z^*)$; indeed the distribution is predicted to be negative between the layers which is clearly physically impossible.

The failure of the truncated distribution is not surprising, for we can see from Table II that the next order parameter, $\langle \tau_2 \rangle$, is not negligible. Indeed, if this is used to extend the expansion to two terms, then $\rho^*(z^*)$ so predicted is found to be in good agreement with the distribution obtained directly from the simulation [see Fig. 10(b)]. The translational order

TABLE II. The scaled layer spacing, d^* , and the translational order parameters, $\langle \tau_n \rangle$, for the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$) and the smectic B phase ($T^* = 1.20$, $\rho^* = 0.2110$) of the mesogen GB(4.4, 20.0, 1, 1).

	d^*	$\langle \tau_1 \rangle$	$\langle \tau_2 \rangle$	$\langle \tau_3 \rangle$	$\langle \tau_4 \rangle$	$\langle \tau_5 \rangle$	$\langle \tau_6 \rangle$
SmA	3.85	0.595	0.189	0.038	0.005	...	...
SmB	3.84	0.842	0.520	0.248	0.094	0.020	0.008

parameters for the smectic B phase are found to be significantly larger than for the smectic A phase. Accordingly, we would expect the retention of just the first nontrivial term in the formal expansion to provide a poor representation of the distribution function. This proves to be the case; in fact, to represent the singlet translational distribution function for the smectic B, it is necessary to retain the first four terms in the expansion.

In view of the need to retain so many terms in the formal expansion of $\rho(z)$, it is clear that an alternative approach is necessary. One possible theoretical treatment is to use the variational analysis to give the singlet translational distribution function. This has been done by Kventzel *et al.*,⁴⁴ who have developed a theory using three order parameters, $\langle P_2 \rangle$, $\langle \tau_1 \rangle$ and $\sigma_{2,1}$, to construct the energetic contribution to the Helmholtz free energy. In their treatment of the problem, they assume that the translational and orientational coordi-

nates are uncorrelated which means that the mixed order parameter, $\sigma_{2,1}$, is the product of the pure order parameters, $\langle P_2 \rangle$ and $\langle \tau_1 \rangle$. This theory then predicts that the singlet translational distribution function has the form

$$\rho(z) = Z_\tau^{-1} \exp(g_1 \cos(2\pi z/d)), \quad (21)$$

where the parameter g_1 depends on both the translational and orientational order parameters and Z_τ is a normalization factor. The same functional form is predicted by the McMillan theory²⁹ but this results from an assumption about the distance dependence of the pair potential used to develop the molecular field theory. As for the singlet orientational distribution function, our primary concern is with the form of $\rho(z)$ rather than the prediction of g_1 . We have, therefore, fitted Eq. (21) to the distribution function determined for the smectic A phase from the simulation. The result is shown in Fig. 11(a) and we can see that the predicted distribution is in complete agreement with the distribution obtained directly

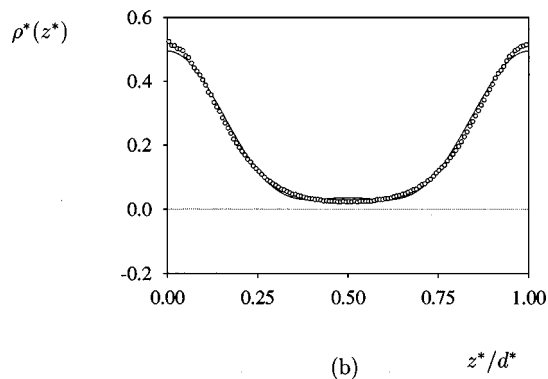
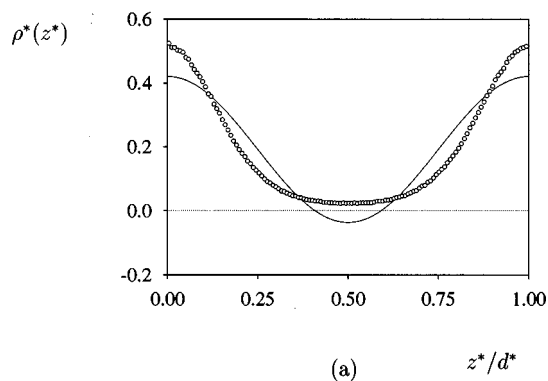


FIG. 10. The dependence of the one dimensional singlet translational distribution function, $\rho^*(z^*)$, on the scaled distance, z^*/d^* , between the center of one layer and that of the next in the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$). The simulation data obtained by averaging over all the layers are shown as points and the predictions of the expansion for $\rho^*(z^*)$ [see Eq. (20)] are shown as solid lines, truncated after (a) one term and (b) two terms.

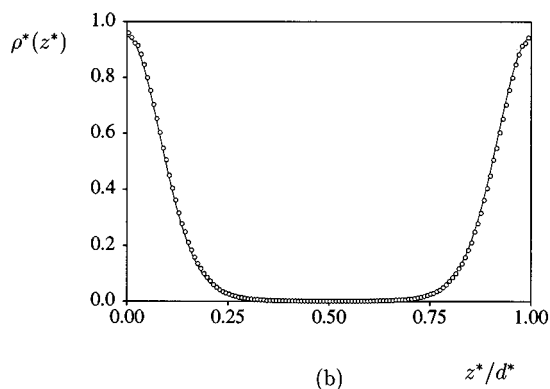
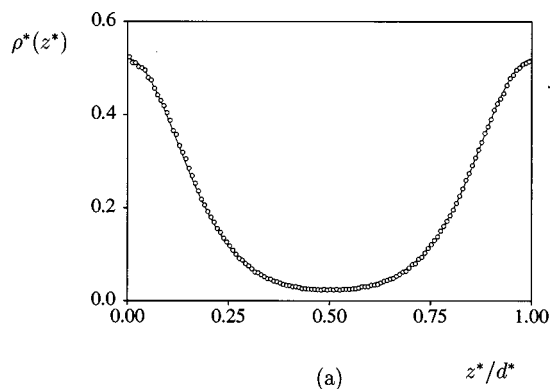


FIG. 11. The comparison of the singlet translational distribution function, $\rho^*(z^*)$, predicted by molecular field theory [see Eq. (21)] and that determined from the simulation of the mesogen GB(4.4,20.0,1,1) in (a) the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$) and (b) the smectic B phase ($T^* = 1.20$, $\rho^* = 0.2110$). The simulation data obtained by averaging over all the layers are shown as points and the predictions as solid lines.

from the simulation. Given this success, we have also fitted the theoretical expression for $\rho(z)$ to the translational distribution function for the more highly ordered smectic B phase. The results are shown in Fig. 11(b) and, as for the smectic A phase, there is perfect agreement between theory and simulation. It would seem, therefore, as if the molecular field theories of the smectic A phase^{29,44} correctly predict the form of the singlet translational distribution function, $\rho(z)$. The reason for their success with the smectic B phase is less clear, given the high translational order of this phase. However, in the limit of high translational order, the same argument which was used to obtain the form of $f(\beta)$ in the high orientational order limit can be applied. This means that $\rho(z)$ takes the general form

$$\rho(z) = Z_{\tau}^{-1} \exp \left(\sum_n g_n \cos(2\pi n z/d) \right), \quad (22)$$

and, in the high order limit, this becomes

$$\rho(z) = Z_{\tau}^{-1} \exp(-\xi_1 z^2/d^2), \quad (23)$$

irrespective of whether the summation is restricted to just one term, as in Eq. (21), or whether they all contribute.

We now turn to the coupling between the orientational and translational coordinates. The singlet orientational and translational distribution function, $P^*(z^*, \beta)$, is shown for the nematic, smectic A and smectic B phases in Fig. 12. In the nematic phase, the position of a particle has no effect on its orientation; each cross section of $P^*(z^*, \beta)$ at a given z^* produces a function equivalent to the singlet orientational distribution function, $f(\beta)$, except for a constant normalization factor. This is as we expect for a phase which has no long range translational order. The periodicity in the number density along z^* becomes apparent for the smectic A phase and it is of interest to determine what effect the position of a particle has on its orientation. Slices through the distribution function at a given z^* reveal that the resulting function is independent of z^* , apart from a normalization factor which is related to the probability of finding a particle with a particular z coordinate along the director. This implies that the orientational distribution is independent of the position of the particle and that there is no coupling between the orientational and translational coordinates. This does not support the assumptions made in the theory proposed by McMillan;³⁰ however, it does support the assumption made by Kvetsel *et al.*,⁴⁴ that the probability of finding a particle with a particular orientation is independent of its translational coordinate. On further cooling to the smectic B phase, there is essentially zero probability that a particle will be found between the layers. However, for all slices at a given z^* with a finite probability, we again find that the form of the orientational distribution function is independent of the position and simply scales with $\rho^*(z^*)$. This behavior again supports the assumption made by Kvetsel *et al.*⁴⁴ for the smectic A phase. We note that this is qualitatively different to the results obtained by van Roij *et al.*⁴⁵ for the smectic A phase exhibited by a system of hard spherocylinders. They found that, within a layer, the orientational distribution was peaked at $\cos \beta$ equal to 1.0 and that all of the particles had an orientation within the cone $0^\circ \leq \beta \leq 41^\circ$. However, for a sec-

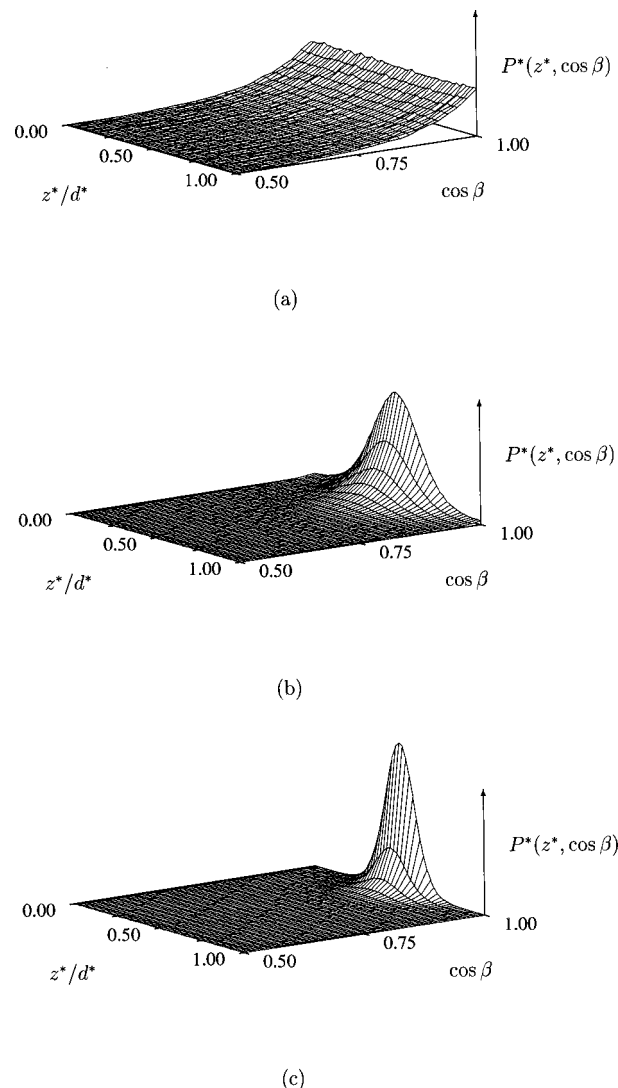


FIG. 12. The singlet orientational-translational distribution function, $P^*(z^*, \cos \beta)$, in (a) the nematic phase ($T^*=1.60$, $\rho^*=0.1756$), (b) the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and (c) the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$). Note that the layer spacing, d^* , for the nematic phase, which is not well-defined for a translationally disordered phase, is arbitrarily taken to be 3.85.

tion taken between the layers, a bimodal distribution was observed, with a second peak in the interval corresponding to angles of $76^\circ \leq \beta \leq 90^\circ$. This second peak indicates the presence of a finite number of particles between the layers, pointing in a direction perpendicular to the director.

The absence of correlations between the orientational and translational coordinates observed for the smectic phases of the mesogen GB(4,4,20,0,1,1) can be explored via the mixed orientational-translational order parameters, $\sigma_{L,n}$, [see Eq. (6)]. In the decoupled limit, this order parameter is given by⁴⁴

$$\sigma_{L,n} = \langle P_L \rangle \langle \tau_n \rangle, \quad (24)$$

and the results of the simulation are in good agreement with this result for both the smectic A and smectic B phases. The values of the pure and mixed order parameters are given in Table III, together with the products $\langle P_L \rangle \langle \tau_n \rangle$; these are seen to be in close agreement with the values of $\sigma_{L,n}$ deter-

TABLE III. The mixed orientational–translational order parameters, $\sigma_{L,n}$, for (a) the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$) and (b) the smectic B phase ($T^* = 1.20$, $\rho^* = 0.2110$) of the mesogen GB(4.4, 20.0, 1, 1). Note that $\sigma_{L,0}$ and $\sigma_{0,n}$ are given by the pure order parameters $\langle P_L \rangle$ and $\langle \tau_n \rangle$, respectively (see Tables I and II). The products $\langle P_L \rangle \langle \tau_n \rangle$, which are the predicted values for $\sigma_{L,n}$ in the absence of orientational and translational correlations [see Eq. (24)], are given in parentheses.

	(a)				(b)				
	$n=0$	$n=1$	$n=2$	$n=3$	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$
$L=0$	1.000	0.595	0.189	0.038	1.000	0.842	0.520	0.248	0.094
$L=2$	0.884	0.522	0.158	0.028	0.946	0.792	0.482	0.222	0.081
		(0.526)	(0.167)	(0.034)		(0.797)	(0.492)	(0.235)	(0.089)
$L=4$	0.668	0.400	0.123	0.022	0.832	0.698	0.426	0.198	0.073
		(0.397)	(0.126)	(0.025)		(0.700)	(0.433)	(0.206)	(0.078)
$L=6$	0.437	0.267	0.084	0.016	0.682	0.574	0.352	0.164	0.061
		(0.258)	(0.082)	(0.017)		(0.574)	(0.355)	(0.169)	(0.064)
$L=8$	0.252	0.157	0.050	0.009	0.524	0.442	0.273	0.129	0.048
		(0.150)	(0.048)	(0.009)		(0.441)	(0.272)	(0.130)	(0.049)

mined directly from the simulation. We conclude that there is no evidence for transverse interlayer particles in the thermotropic smectic phases exhibited by our mesogenic GB(4.4,20.0,1,1) system. Indeed, there appears to be no coupling between the translational and orientational coordinates at all, as assumed by Kvetsel *et al.*⁴⁴ Whether this is the case for real systems, in which the alkyl chains between the smectic layers are likely to be less ordered than the rigid

units within the layers, is unclear and cannot, of course, be tested using simulations of such single site models.

The single particle properties which we have discussed so far clearly provide important information on the long range structure of the phases. We now turn to the pair distribution functions and correlation coefficients, which are able to give details of the structure at both short and long range. We begin our discussions with the radial distribution func-

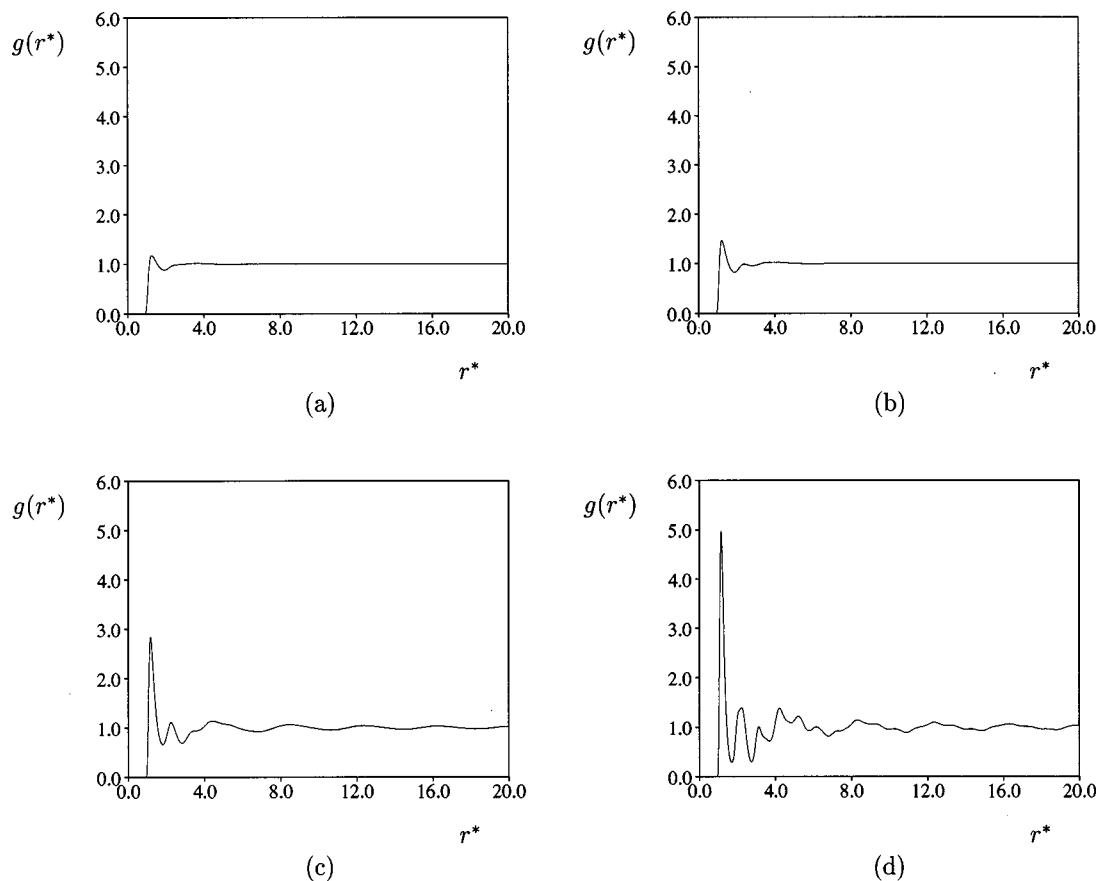


FIG. 13. The radial distribution function, $g(r^*)$, for the mesogen GB(4.4,20.0,1,1) in (a) the isotropic phase ($T^* = 1.80$, $\rho^* = 0.1562$), (b) the nematic phase ($T^* = 1.60$, $\rho^* = 0.1756$), (c) the smectic A phase ($T^* = 1.40$, $\rho^* = 0.1932$) and (d) the smectic B phase ($T^* = 1.20$, $\rho^* = 0.2110$).

tion, $g(r)$, which is shown in Fig. 13 for the four phases studied. In the isotropic and nematic phases, the only structure in the radial distribution function is at distances corresponding to the dimensions of the Gay-Berne particles. This clearly indicates the lack of long range translational order, so confirming the assignment of the phase at T^* of 1.60 as a nematic. There is a remarkably small change in $g(r^*)$ on going from the isotropic to the nematic phase, even though the temperatures and densities at which they were studied differ significantly. This suggests that the local structure in the isotropic and nematic phases are very similar, in agreement with x-ray scattering experiments.³³ There is a relatively small increase in the intensity of the peak at a separation of just greater than one, which is presumably associated with the growth in the local orientational order, which allows particles to pack more effectively in the side-by-side configuration. In the smectic A phase, this nearest neighbors peak is quite pronounced [see Fig. 13(c)], as we might expect from the packing of particles in layers. The shell of next nearest neighbors is also apparent from the peak at a scaled separation of about 2.0. However, the new feature in $g(r^*)$, at least in comparison with that found for the nematic phase, is a slight oscillation at long range with a periodicity of about 4.0. This is clearly associated with the layer structure in the smectic A phase, although the small amplitude of the oscillations results from the radial nature of this distribution function. In the smectic B phase, the peaks in $g(r^*)$ have become sharper and a fine structure has appeared on the long range oscillations [see Fig. 13(d)]. The scaled separation between the peaks constituting the fine structure is about one and so we attribute this to the hexagonal packing within the smectic layers. This is also responsible for the poorly resolved splitting on the peak at r^* of about 2.0.

Although the radial distribution function clearly provides some structural information about the two smectic phases, their one dimensional translational order means that the radial character of this distribution is not really appropriate to explore the structure of these phases. Clearly the longitudinal and in-plane distribution functions, defined in the previous section, are more appropriate and we now consider these for the smectic A and smectic B phases. We start with the longitudinal distribution function, $g_{\parallel}(r_{\parallel}^*)$, and, as expected, this function is featureless in the nematic phase, indicating its translationally disordered nature along the director. In marked contrast, the periodic nature of $g_{\parallel}(r_{\parallel}^*)$ in the smectic A phase, shown in Fig. 14(a), indicates the tendency for the particles to arrange themselves in layers. The average separation between the peaks can be used to determine the scaled layer spacing, d^* ; this is found to be 3.85, in agreement with the layer spacing determined from the box dimensions (see Table II). On cooling to the smectic B phase, the peaks become both sharper and more intense, in keeping with the higher translational order in this phase. However, we find that the layer spacing is virtually unchanged at 3.84; indeed, the layer spacing apparently shows no temperature dependence in either of the smectic phases. This layer spacing corresponds to $0.875l$, where l is the particle length, which is similar to that of $0.867l$ found for the mesogen GB(3.0,5.0,2,1).⁹ If we consider the high orientational order

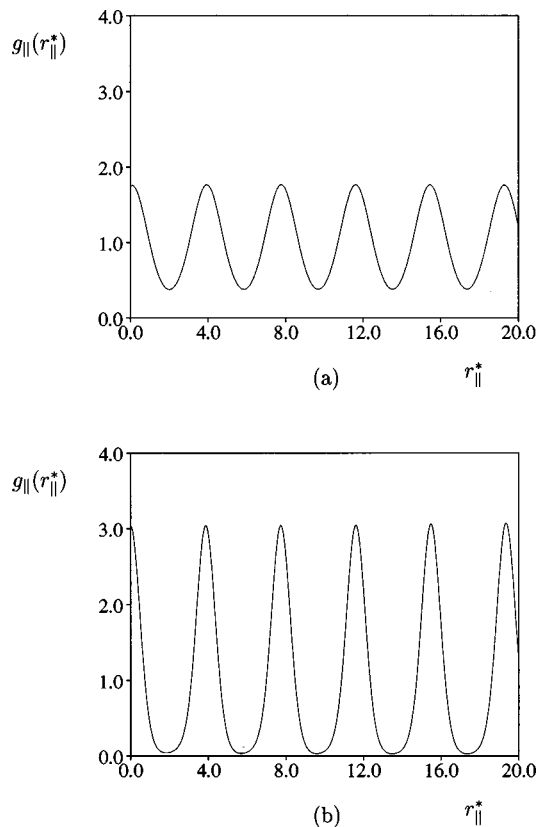


FIG. 14. The longitudinal distribution function, $g_{\parallel}(r_{\parallel}^*)$, for the mesogen GB(4.4,20.0,1,1) in (a) the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and (b) the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$).

in this phase, it is clear that this difference between the layer spacing and the particle length cannot originate from the orientational fluctuations of the particles, as sometimes suggested for real smectic phases.³³ In fact, the difference here is due only to the interdigitation of neighboring layers, which can be related to the ellipsoidal shape of the Gay-Berne particles.

We note that, while similar in form, the pair distribution function, $g_{\parallel}(r_{\parallel})$, is not the same as the single particle distribution, $\rho(z)$, although these are clearly related, becoming equal in the limit of perfect translational order. The latter gives the probability of finding a particle at a position along the director measured from an origin normally taken to be the center of a smectic layer whilst the former gives the probability of finding a particle at a resolved separation along the director from another. We expect, therefore, that the intensity of the peaks in the single particle distribution will not change with location, which is what we observe (see Fig. 9), whereas those for the pair distribution might be expected to decay with increasing resolved separation between the particles, as the translational correlations are lost. For the smectic phases of the mesogen GB(4.4,20.0,1,1), this is not observed, showing that the correlation length must be greater than half the box dimension along the z axis of the system which we have studied.

The translational order within the smectic layers has been characterized by the in-plane distribution function, $g_{\perp}(r_{\perp}^*)$, with the results shown in Fig. 15. In the smectic A

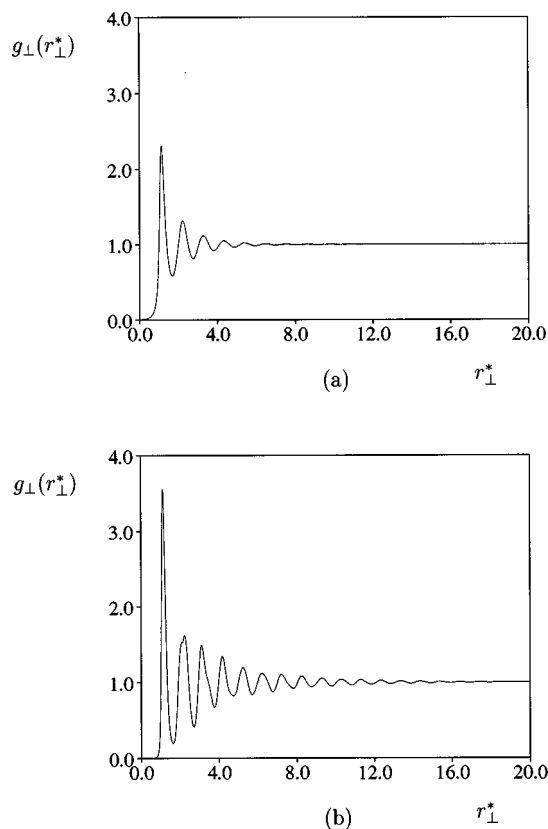


FIG. 15. The in-plane distribution function, $g_{\perp}(r_{\perp}^*)$, for the mesogen GB(4,4,20,0,1,1) in (a) the smectic A phase ($T^*=1.40$, $\rho^*=0.1932$) and (b) the smectic B phase ($T^*=1.20$, $\rho^*=0.2110$).

phase, it is apparent that the in-plane order is liquidlike; there is no long range structure in $g_{\perp}(r_{\perp}^*)$ above $r_{\perp}^*$ of 6.0. However, the small number of well-resolved peaks with decreasing intensity indicates that there is short range translational order within the layers. This in-plane order extends to larger separations in the smectic B phase, although again $g_{\perp}(r_{\perp}^*)$ tends to unity for large $r_{\perp}^*$, indicating that the translational order is not long range within the layers. We observe that the first nearest neighbors peak occurs at scaled separations, $r_{\perp}^*$, of 1.15 and 1.14 in the smectic A and smectic B phases, respectively. The particles, therefore, tend to arrange themselves at separations corresponding to the potential well minimum for pairs in the side-by-side configuration, as expected. We also observe that the first peak in $g_{\perp}(r_{\perp}^*)$ for the smectic B phase is narrower and more intense than that in the smectic A, indicating that the nearest neighbor cluster is more ordered at the lower temperature. The structure of the in-plane distribution is qualitatively different in the two phases for the next nearest neighbors. Thus in the smectic A phase, the second peak is rather weak and broad in comparison to the first. In the smectic B phase, the distribution at $r_{\perp}^*$ between 2.0–2.5 clearly shows two overlapping peaks, although they are not well resolved. It is these peaks which are characteristic of a hexagonal in-plane arrangement; thus, the short range translational order in this phase is, on average, hexagonal. This prompts us to conclude that this phase is most probably a smectic B phase, rather than an ordered smectic A.

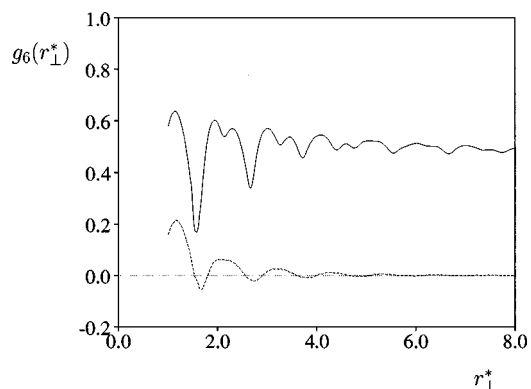


FIG. 16. The bond orientational correlation function, $g_6(r_{\perp}^*)$, for the mesogen GB(4,4,20,0,1,1) in the smectic A phase (— —) and the smectic B phase (—).

This assignment of the low temperature smectic phase as a SmB can be confirmed by examining the in-plane bond orientational order, through the distribution function $g_6(r_{\perp}^*)$; this function is shown in Fig. 16 for the two smectic phases. It is apparent that $g_6(r_{\perp}^*)$ quickly tends to zero in the smectic A phase and, therefore, that there is no long range hexatic or bond orientational order in this phase. Conversely, in the smectic B phase, the bond orientational correlation function tends to a finite value, which is indicative of a phase with hexatic order. The minima in $g_6(r_{\perp}^*)$, observed for both smectic A and B phases, are found to occur at in-plane distances between shells of neighbors and result from the small number of particles found at these separations. It is on the basis of the two in-plane functions, $g_{\perp}(r_{\perp})$ and $g_6(r_{\perp})$, that we classify the low temperature phase as a smectic B; these functions clearly indicate that the phase has only short range translational order but long range bond orientational order.

The long range orientational order of the various phases has been established via the second rank orientational order parameter. We now explore the short range orientational correlations, with the aid of the particle frame orientational correlation coefficients, $g_{L_i L_j m}(r)$, which were described in Sec. IV. We have evaluated these in the isotropic and nematic phases for L_i and L_j up to a maximum value of four but here we deal with the dominant coefficients, namely $g_{020}(r)$, $g_{220}(r)$, $g_{221}(r)$, and $g_{222}(r)$. The dependence of this complete set of second rank coefficients on the scaled separation, r^* , is shown in Fig. 17. Before discussing these results, we note that in the isotropic phase they have a structure which is analogous to that found for hard ellipsoids with κ equal to five,⁴⁶ although the structure is not quite so pronounced, presumably because of the slightly lower density used in the simulation. This similarity is perhaps to be expected, given the ellipsoidal shape of the Gay–Berne particles. As we have seen, the long range limit for $g_{LLm}(r^*)$ in an orientationally ordered phase is $(-)^m \langle P_L \rangle^2$ and our results for the nematic phase are in accord with this, using the value for $\langle P_2 \rangle$ of 0.69 calculated directly. The remaining coefficient, $g_{020}(r^*)$, decays to zero, as expected. All of the orientational correlation coefficients decay to zero in the isotropic phase, in accordance with the absence of long range orientational order. The decay to the long range limit is complete within a scaled

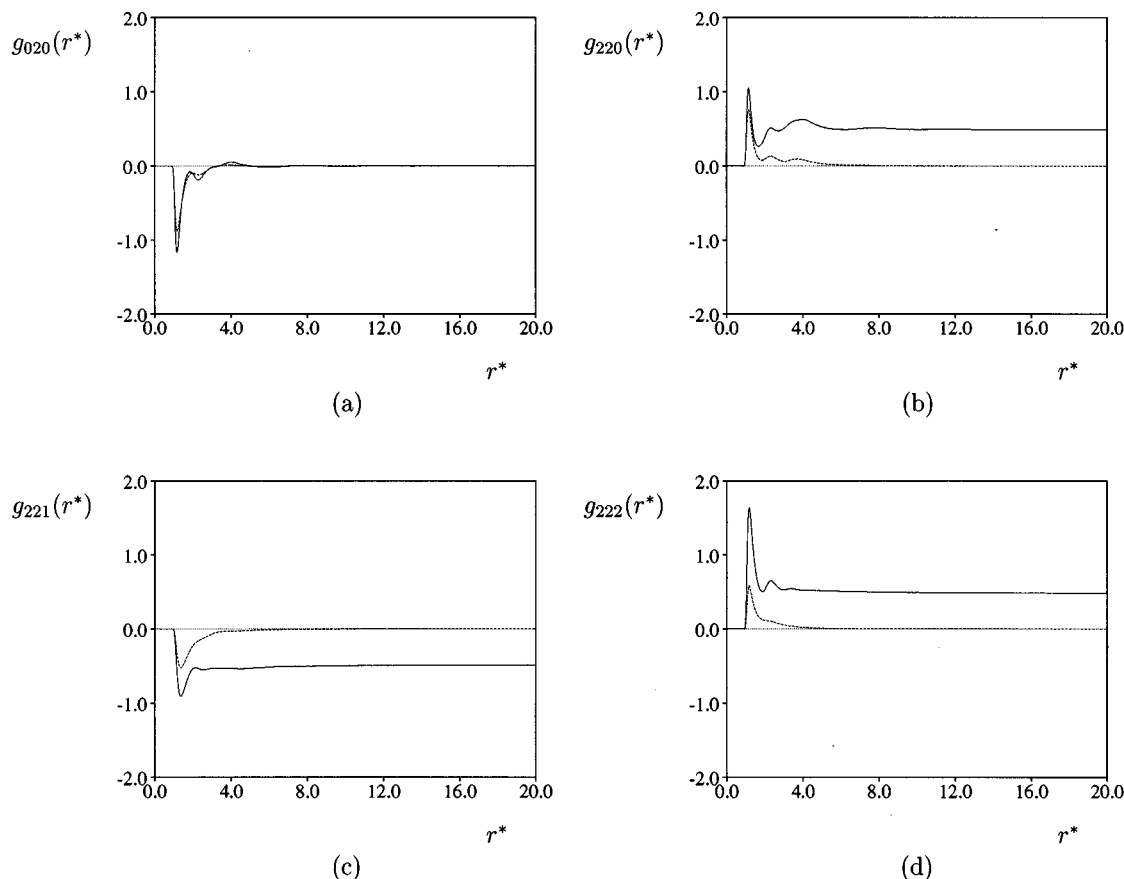


FIG. 17. The dependence of the orientational correlation coefficients, $g_{L_i L_j m}(r^*)$, calculated in the particle frame on the scaled distance, r^* , for the mesogen GB(4.4,20.0,1,1) in the isotropic phase (---) and the nematic phase (—).

radial separation, r^* , of ~ 6.0 , which shows how rapidly the direct correlations are lost. Some of the structure in these coefficients at short range can be understood in relatively simple geometric terms, as we shall illustrate. To do this, we need explicit expressions for the spherical harmonics in terms of the spherical polar angles ($\beta\alpha$). These give the orientational correlation coefficients, which we show in Fig. 17, as

$$g_{020}(r) = \frac{\sqrt{5}}{2} g(r) \langle (3 \cos^2 \beta_j - 1) \rangle_r,$$

$$g_{220}(r) = \frac{5}{4} g(r) \langle (3 \cos^2 \beta_i - 1)(3 \cos^2 \beta_j - 1) \rangle_r, \quad (25)$$

$$g_{221}(r) = \frac{15}{2} g(r) \langle \sin \beta_i \cos \beta_i \sin \beta_j \cos \beta_j \exp i(\alpha_i - \alpha_j) \rangle_r,$$

and

$$g_{222}(r) = \frac{15}{8} g(r) \langle \sin^2 \beta_i \sin^2 \beta_j \exp 2i(\alpha_i - \alpha_j) \rangle_r.$$

For particles at a scaled separation of about unity, we expect that the repulsive forces will tend to make the particles orthogonal to the interparticle vector. This means that β_i and β_j will approximate to 90° , although the azimuthal angles could take a range of values; however, for nearest neighbors, especially in the nematic phase, it is expected that the particles will be parallel (see Fig. 7). The other quantity needed to evaluate the correlation coefficients is the radial distribution function (see Fig. 13). In the isotropic phase, $g(r^*)$ is close to unity, except for r^* less than one and, even in the

nematic phase, it is only the first peak in $g(r^*)$ which deviates to any extent from unity. With these assumptions, the correlation coefficient $g_{020}(r^*)$ for nearest neighbors should be about -1.1 , which is close to the value found in the nematic phase and slightly greater than that in the isotropic phase. This reduction in $g_{020}(r^*)$ presumably results from deviations of β_j from 90° , caused by the ellipsoidal shape of the particles. There is a much weaker negative peak at r^* of about 2, which comes from particles in the next shell of neighbors but there will also be contributions from neighboring particles which are not side-by-side. The next correlation coefficient is $g_{220}(r^*)$ and, if both particles are orthogonal to the interparticle vector, then this coefficient should be about 1.25 for nearest neighbors. This prediction is somewhat larger than the observed value in the isotropic phase, whilst in the nematic phase the agreement is quite good. The interpretation of the peaks at larger radial separations is more problematic because, as we have seen, different configurations of the particles can contribute at these separations. The coefficient $g_{222}(r^*)$ for nearest neighbors is predicted to have a value of about 1.9, provided the particles are parallel; the dependence on the difference in the azimuthal angle will reduce this if the particles are not parallel. In the nematic phase, $g_{222}(r^*)$ is found to be in good agreement with this prediction. The value of $g_{222}(r^*)$ is significantly reduced in the isotropic phase because the particles do not retain their parallel alignment. This simple geometric argument appears

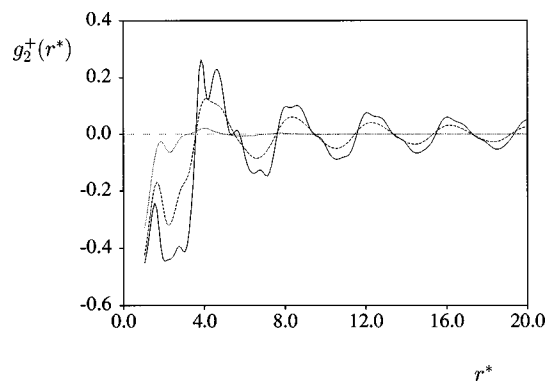


FIG. 18. The second rank orientational correlation coefficient, $g_2^+(r^*)$, which gives the order parameter for the interparticle vector as a function of its length for the mesogen GB(4.4,20.0,1,1) in the nematic phase ($\cdots$), the smectic A phase ($---$), and the smectic B phase ($—$).

to work quite well for these correlation coefficients but it seems to fail for $g_{221}(r^*)$. Here, for nearest neighbors orthogonal to the interparticle vector, this should vanish but, as we can see in Fig. 17(c), it is just less than unity in the nematic phase and significantly less in the isotropic phase as found for hard ellipsoids.⁴⁶ It may be significant, however, that the extremum occurs at a slightly larger separation than for the other orientational correlation coefficients. This may indicate that the particles contributing to $g_{221}(r^*)$ deviate from the side-by-side configuration and so the angles β_i and β_j would be less than 90° , thus accounting for the nonzero values of this correlation coefficient at small separations, a result enhanced by the large numerical factor multiplying the angular function [see Eq. (25)]. Before leaving the orientational correlation coefficients we should note that they enter the scheme for the calculation of the direct pair correlation function; in turn, this function is important in many theoretical calculations, such as that for the elastic constants of a nematic liquid crystal.⁴⁷ The expansion in the particle frame is only useful to describe translationally disordered phases; the averaging in spherical shells is not appropriate to characterize, for example, the partial translational order in smectic phases.

We now describe the final orientational correlation coefficient, $g_2^+(r^*)$ [see Eq. (9)], which gives the distance dependence of the second rank orientational order parameter for the interparticle vector with respect to the director. The results for the nematic, smectic A and smectic B phases are shown in Fig. 18. Within the nematic phase, $g_2^+(r^*)$ adopts a large negative value for particles at a scaled separation of approximately unity; this is to be expected because such pairs of particles should be in a side-to-side configuration and, since the particles tend to be parallel to the director, the associated interparticle vector should be orthogonal to the director, thus giving a negative value for $\langle P_2^+(\cos \beta_r) \rangle_{r^*}$.³⁹ The value for $g_2^+(r^*)$ at this separation is predicted to be about $-\frac{1}{2}\langle P_2 \rangle$, which is -0.35 for this system. This is essentially identical to the value observed (see Fig. 18). However, for larger separations, $g_2^+(r^*)$ rapidly decays to zero, indicating that the interparticle vector is uniformly distributed, thus supporting an assumption often made in molecular

field theories of nematics.²⁸ Within the smectic phases, $g_2^+(r^*)$ is slightly less than -0.5 at a scaled separation of approximately unity, which is to be anticipated given the high orientational order of these phases. Unlike the nematic phase, $g_2^+(r^*)$ for the smectic phases does not decay to zero but oscillates with a periodicity equal to the layer spacing. This is also to be expected, given the layered structure of these phases; the negative values of $g_2^+(r^*)$ come from pairs of particles within the same layer and the positive values originate from pairs of particles in different layers. In the smectic B phase, there is an additional oscillation on $g_2^+(r^*)$, with a periodicity of about unity, originating from the local hexagonal structure of the phase.

VI. SUMMARY AND CONCLUSIONS

We have investigated the behavior of a Gay-Berne liquid crystal with the four parameters characterizing the potential chosen to have values similar to those expected for real mesogenic molecules. Isothermal-isobaric Monte Carlo simulations have shown that the mesogen GB(4.4,20.0,1,1) exhibits isotropic, nematic, smectic A and smectic B phases, depending on the choice of the pressure. We have determined the phase diagram for this mesogen and have found that it is in good agreement with those found for real mesogens. This parametrization of the potential also seems to predict reasonably accurately the transitional properties, with the exception of the fractional volume change at the nematic-isotropic transition. Here, the Gay-Berne potential gives a value approximately ten times larger than that observed experimentally. It seems likely that this failure of the model potential results from the ellipsoidal shape of a particle.

The structures of the phases have been investigated using canonical molecular dynamics simulations of a much larger system containing 16 000 particles. This has allowed us to investigate the correlations, both orientational and translational, over relatively large distances. In addition, the singlet orientational and translational distribution functions have been calculated. The orientational distribution function, $f(\beta)$, for the nematic phase is found to be in essentially complete agreement with that predicted by the Maier-Saupe theory. However, this should not be taken as support for the original assumptions on which the theory is based but rather, according to the variational principle, it is a consequence of the dominance of the second rank orientational order parameter. The same Maier-Saupe distribution function is also in good agreement with our results for the smectic A and smectic B phases. The success of the theory here is attributed to the high orientational order of these phases, which means that the distribution function is essentially Gaussian. The singlet translational distribution function, $p(z)$, has been evaluated for both smectic phases. In the smectic A phase, the distribution is found to approximate to a simple sinusoidal form; however, this fails completely in the smectic B phase. We have found that the McMillan distribution gives an essentially exact account of our results for both the smectic A and smectic B phases. In the former, this almost certainly results from the dominance of the first translational order

parameter, while in the latter, this comes from the high translational order of the phase. The layer spacing shows that there is significant interdigitation of the layers which is attributed to the ellipsoidal shape of the Gay–Berne particles. The mixed orientational–translational distribution function reveals that the translational and orientational coordinates in both the smectic A and smectic B phases are uncorrelated. This supports the assumption made by Kventzel, Luckhurst, and Zewdie in their simplification of the McMillan theory of the smectic A phase. It does, however, contrast with the results found for hard spherocylinders, where there is a strong dependence of the orientational distribution on the position of the particles in the smectic layers.

The radial distribution functions determined for the isotropic and nematic phases are very similar, which suggests that the local structure in the two phases are also similar, in agreement with x-ray scattering studies of real systems. The layer structure of the smectic phases is best revealed by the longitudinal distribution function. The additional structure within a single layer can be explored with the in-plane distribution function. For the smectic A phase, $g_{\perp}(r_{\perp})$ exhibits some structure but at short range only, as expected for this phase. Within the layers of the smectic B phase, the range of the translational correlations is found to grow but not to extend to long range. The essential difference between the smectic A and smectic B phases is revealed by the bond orientation correlation function which decays to zero for the smectic A but not for the smectic B. The orientational correlations in the isotropic and nematic phases have been probed with the second rank, particle frame correlation coefficients. These also reveal the similarity in the local structure at short range for the two phases. At long range, the coefficients in the isotropic phase decay to zero but those in the nematic phase do not because of the long range orientational order. The behavior of the $g_{22m}(r)$ at very short range is in accord with the expected local structure but for scaled separations greater than about two, the interpretation is far more complex because of the many contributions to the correlation coefficients. The distribution of the interparticle vector in the nematic and the two smectic phases has been explored with the distance dependent second rank order parameter. For the nematic phase, this order parameter decays rapidly to zero but, at short range, there are significant deviations from zero. This long range behavior is in accord with the assumptions made in molecular field theories that the interparticle is distributed uniformly. The short range structure in $g_2^+(r)$

clearly reveals the failure of this assumption and is of particular significance for the contributions which electrostatic interactions make to the stability of the nematic phase. The long range anisotropic nature of the translational structure of the smectic phases means that the interparticle order parameter should deviate from zero even at long range, as we find.

It is clear that the Gay–Berne potential with parameters selected to be appropriate for mesogenic molecules provides a good model for thermotropic liquid crystals. The phases formed, their structures and the transitional properties are found to be in close accord with the behavior of real mesogens. Such a system, therefore, allows us to explore in a general manner the influence of particular molecular interactions on liquid crystal characteristics. In addition, the results of these simulations of the mesogen GB(4.4,20.0,1,1) provide a valuable test bed for molecular theories of liquid crystals formed from rodlike particles.

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APPENDIX A

The Gay–Berne potential^{4,7} has a familiar 12-6 dependence on the shifted and scaled separation, R , between the uniaxial particles

$$U(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \mathbf{r}) = 4\epsilon(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \mathbf{r})(R^{-12} - R^{-6}), \quad (\text{A1})$$

where

$$R = (r - \sigma(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \mathbf{r}) + \sigma_0)/\sigma_0. \quad (\text{A2})$$

The orientations of the symmetry axes of the particles i and j are denoted by the unit vectors $\hat{\mathbf{u}}_i$ and $\hat{\mathbf{u}}_j$ while the orientation of the vector joining them is denoted by $\mathbf{r}$. The dependence of the contact distance on the orientations of the particles and the interparticle vector is

$$\sigma(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \mathbf{r}) = \sigma_0 \left[1 - \chi \left(\frac{(\hat{\mathbf{u}}_i \cdot \mathbf{r})^2 + (\hat{\mathbf{u}}_j \cdot \mathbf{r})^2 - 2\chi(\hat{\mathbf{u}}_i \cdot \mathbf{r})(\hat{\mathbf{u}}_j \cdot \mathbf{r})(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)}{1 - \chi^2(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2} \right) \right]^{-1/2}, \quad (\text{A3})$$

where σ_0 is the contact distance for the cross configuration ($\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j = \hat{\mathbf{u}}_i \cdot \mathbf{r} = \hat{\mathbf{u}}_j \cdot \mathbf{r} = 0$). The parameter χ is a function of the ratio $\kappa (\equiv \sigma_e/\sigma_s)$, which is defined in terms of the contact distances when the particles are end-to-end (e) and side-by-side (s),

$$\chi = \frac{\kappa^2 - 1}{\kappa^2 + 1}. \quad (\text{A4})$$

This vanishes for a sphere and tends to the limiting value of

unity for an infinitely long rod. The orientational dependence of the potential well depth is given by a product of two functions,

$$\epsilon(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \hat{\mathbf{r}}) = \epsilon_0 \epsilon''(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) \epsilon'(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \hat{\mathbf{r}}), \quad (\text{A5})$$

where the scaling parameter ϵ_0 is the well depth for the cross configuration. The first of these functions

$$\epsilon(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j) = [1 - \chi^2(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2]^{-1/2}, \quad (\text{A6})$$

clearly favors the parallel alignment of the particles and so aids liquid crystal formation. The second function has a form analogous to $\sigma(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \hat{\mathbf{r}})$ and so is somewhat more involved; it is

$$\begin{aligned} \epsilon'(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j, \hat{\mathbf{r}}) \\ = 1 - \chi' \left(\frac{(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}})^2 + (\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}})^2 - 2\chi'(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{r}})(\hat{\mathbf{u}}_j \cdot \hat{\mathbf{r}})(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)}{1 - \chi'^2(\hat{\mathbf{u}}_i \cdot \hat{\mathbf{u}}_j)^2} \right), \end{aligned} \quad (\text{A7})$$

where the parameter χ' is determined by the ratio of the well depths, $\kappa' (\equiv \epsilon_s / \epsilon_e)$, via

$$\chi' = \frac{\kappa'^{1/\mu} - 1}{\kappa'^{1/\mu} + 1}. \quad (\text{A8})$$

This second function favors the side-by-side arrangement over the end-to-end and so encourages the formation of smectic phases.

Just as there are many different compounds which form liquid crystal phases, so there are many different parametrizations of the Gay–Berne potential which exhibit liquid crystal behavior. It is necessary, therefore, to have a systematic notation which distinguishes between the different sets of parameters and we suggest GB($\kappa, \kappa', \mu, \nu$). This notation could also be used to differentiate between rods and discs provided that κ is defined as the ratio of the contact distances for discs in the face-to-face (f) and edge-to-edge (e) arrangements, that is, $\kappa \equiv \sigma_f / \sigma_e$. The ratio κ will, therefore, be greater than unity for rods and less than unity for discs. The definitions of κ' for rods and discs are however different, the convention being that this ratio should be greater than unity. Thus, normally for rods $\kappa' \equiv \epsilon_s / \epsilon_e$, while for discs it is more sensibly defined as $\kappa' \equiv \epsilon_f / \epsilon_e$. There is still an ambiguity about the choice of the parameter entering the scaled and shifted distance, R , [see Eq. (A2)] in the Gay–Berne potential. For rods, this is defined as the contact distance when the particles are in the cross configuration. Use of the same configuration for discs means that this would be identified as σ_e but this makes the potential tend to infinity only at negative separations for all but the edge-to-edge arrangement.⁴⁸ This is clearly unrealistic physically and it has been suggested that σ_f provides a more reasonable choice in the definition of R .⁴⁸ We propose, therefore, that the contact distance entering the definition of R should be σ_s for rods and σ_f for discs but that the distance scaling parameter σ_0 , defined as the contact distance for particles in the cross configuration, be retained for both rods and discs.

APPENDIX B

The even Legendre polynomials used to characterize the orientational order of a liquid crystal via the order parameters and in the singlet orientational distribution function are

$$P_0(\cos \beta) = 1, \quad (\text{B1})$$

$$P_2(\cos \beta) = \frac{1}{2}(3 \cos^2 \beta - 1), \quad (\text{B2})$$

$$P_4(\cos \beta) = \frac{1}{8}(35 \cos^4 \beta - 30 \cos^2 \beta + 3), \quad (\text{B3})$$

$$P_6(\cos \beta) = \frac{1}{16}(231 \cos^6 \beta - 315 \cos^4 \beta + 105 \cos^2 \beta - 5), \quad (\text{B4})$$

$$P_8(\cos \beta) = \frac{1}{128}(6435 \cos^8 \beta - 12\,012 \cos^6 \beta + 6930 \cos^4 \beta - 1260 \cos^2 \beta + 35). \quad (\text{B5})$$

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